

Financialization, Intraday Institutional Trading, and Market Quality

by

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Abstract

We investigate the impact on market quality of an important, albeit hitherto unexplored, aspect of “financialization” – the massive growth in *intraday* positions and short-horizon trading of “non-commercial” institutional financial traders. Our results are based on a comprehensive, non-public regulatory dataset of trades in crude oil futures – the world’s largest commodity futures market. First, we exploit the September 2006 introduction of electronic trading in crude oil futures as an instrument to identify the causal effect of institutional financial traders on key metrics of market quality. We document a sharp increase in trading by institutional financial traders following the direct market access enabled by electronic trading. We exploit large differences in the growth rates of institutional financial trading in contracts with different maturities to tease out the respective roles of electronification and of financialization. A two-stage regression analysis establishes that there is a significant increase in market quality due to financialization. Second, having established causality, we apply structural VAR analysis to further examine the endogenous relationship between the participation of institutional financial traders and market quality metrics between April 2007 and May 2008. We show that an increase in the participation of institutional financial traders contemporaneously reduces the variance of pricing errors, narrows bid-ask spreads, cuts customer trade imbalances, and increases market depth.

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Financialization, Intraday Institutional Trading, and Market Quality

I. Introduction

In the early 2000's, institutional financial actors, driven purely by “non-commercial” financial motives, started assuming a sharply greater role in commodity markets. This development, known as the “financialization” of commodities (UNCTAD, 2011, p 13), has spawned a substantial amount of academic research. Much of the focus to date, however, has been on positions held overnight and relatively “longer-term” trading strategies of commodity index traders (CITs) and managed money traders, and on their relevance to the daily, weekly, or monthly distributions of commodity returns: risk premia (e.g., Acharya, Lochstoer, and Ramadorai, 2013; Singleton, 2014), price levels (e.g., Henderson, Pearson, and Wang, 2014; Hamilton and Wu, 2015), volatility (e.g., Brunetti, Büyüksahin, and Harris, 2016), and co-movements with other asset markets (e.g., Tang and Xiong, 2012; Büyüksahin and Robe, 2014; Cheng, Kirilenko, and Xiong, 2015; Başak and Pavlova, 2016).

However, surprisingly little attention has been paid to another major manifestation of financialization – the massive growth in short-horizon trading of institutional financial traders, with positions held *intraday*. This growth potentially reflects the significantly greater role of institutional financial traders in intraday price formation and liquidity provision. This paper is the first to empirically investigate the impact of this aspect of financialization, and in particular, of the growth in intraday trading of institutional financial traders, on the quality of commodity markets. Unless otherwise indicated, our use of the term “financialization” in this paper is intended to signify the growth in *intraday* trading and positions of institutional financial traders. Our results are based on a comprehensive, non-public regulatory dataset of trading at the world's largest commodity market – the New York Mercantile Exchange (NYMEX) West Texas Intermediate (WTI) sweet crude oil futures market. We use the September 2006 introduction of electronic trading in this market as an

instrument to extract the component of institutional financial trading that is exogenous to market conditions. We are able to distinguish between the effects of electronification and financialization to document a causal link between financialization – changes in the trading activity of institutional financial traders – and changes in intraday pricing efficiency and liquidity.

Specifically, we test hypotheses on the impact of financialization on intraday pricing efficiency by examining the variance of “pricing errors”, i.e., the deviations of market prices from informationally efficient “fundamental” values (Hasbrouck, 1993; Fotak, Raman, and Yadav, 2014). We also test hypotheses on the impact of financialization on intraday liquidity by examining three commonly used measures: bid-ask spread, the Amihud measure of depth, and the absolute magnitude of trade imbalances (i.e., the difference between customer buys and customer sells). Overall, we find unequivocal and strong evidence that institutional financial traders have a significantly beneficial impact on each of these aforesaid measures of market quality.

The NYMEX market for crude oil futures underwent a major structural change with electronification, i.e., the start of electronic trading on Globex on September 5, 2006. Before electronification, trading was physically confined to the pits in Chicago, and intermediated largely by “Locals” – individual traders functioning as scalpers (Garbade and Silber, 1984), and as the main providers of immediacy/liquidity in these markets (Manaster and Mann, 1996). After electronification, electronic trading on Globex started coexisting alongside face-to-face trading in the pits. Importantly, this also resulted in a transformative change in ease of access for traders without physical access to the trading pits, with any such trader being able to trade directly on the electronic system, and thereby compete with Locals in the supply of liquidity through the posting of limit orders to buy and sell, or compete otherwise: for example, through short-horizon trading on information in the order flow. Not surprisingly, consistent with the NASDAQ and London Stock Exchange evidence

(in Barclay et al. (2000) and Naik and Yadav (2003) respectively) for 1997 structural market reforms that allowed public traders to compete with traditional market makers in supplying liquidity, and consistent with electronification-specific equity market evidence in Jain (2005), we find that electronification results in a significant increase in pricing efficiency and a significant reduction in trading costs. That said, this paper's focus is on financialization and not on electronification.

We identify institutional financial trading – the underlying basis for financialization – through a confidential trader-level intraday dataset from the U.S. Commodity Futures Trading Commission (CFTC). This regulatory dataset classifies the traders on each side of a transaction into one of four customer types via an industry-standard Customer Type Indicator (CTI), which ranges from 1 to 4 as follows: (a) CTI-1: trade initiated and executed by *individual* exchange members for accounts they own or control, labelled by us both (pre- and post- electronification) as trading by Locals; (b) CTI-2: trades by institutional exchange members for accounts they own or control; (c) CTI-3: trades by individual exchange members on behalf of other individual members; and (d) CTI-4: non-member customer transactions, i.e., trades by the customers of the exchange or external traders.

The trader category that is of primary interest to us in this paper is the CTI-2 category. In line with the Chicago Mercantile Exchange (CME), we proxy for the participation of (intraday focused) institutional financial traders by these CTI-2 traders, whose trading activity is significant enough to warrant corporate exchange membership for their proprietary trading desks. Corporate exchange membership allows firms to obtain preferential clearing and Globex platform fees, and other significant benefits on their proprietary trading of CME products for both electronic and floor trades. CTI-2 traders can potentially include hedge funds, commodity pools, banks, futures commission

merchants, commodity trading advisors, foreign and domestic broker/dealers, introducing brokers, proprietary trading firms and other eligible corporate entities.²

Given that our financialization focus is on *intraday* positions and the short-horizon trading of institutional financial traders, we note that post-electronification: (a) the median level of hourly inventory turnover of this CTI-2 group is as high as 92% (as against 88% for Locals and 0% for traders in the customer group), and (b) this CTI-2 group has largely displaced the erstwhile Locals as liquidity providers in the most liquid futures contracts. We utilize the growth in the trading activity of these CTI-2 institutional exchange members to proxy for financialization.

It is not surprising that electronification resulted in a sharp increase in the volume of trading by institutional financial traders, and in their aggregate share of total trading volume relative to Locals. However, what is particularly relevant from our perspective is that this financialization – entry of new institutional financial traders – took place, both before and after electronification, largely in the first two front contract months that generally account for most directional trading; and in the next two December months that, together with the two front months, generally account for most of the calendar spread trading. For each trading day, we label the aforesaid four contract month contracts as “short-term” contracts in this paper; and all other contracts as “long-term”. The significant differences in the growth rates of institutional financial trading at different points of the futures term structure enable us to distinguish between the effects of financialization and electronification separately. Using data from January 2006 to March 2007, we extend the two-stage regression methodology of Hendershott, Jones and Menkveld’s (2011) to show that key measures of market quality (bid-ask spreads, market depth, magnitude of customer trade imbalances) and pricing efficiency (variance of pricing errors) all improved significantly due to financialization.

² CTI-2 traders can also be commercial entities, but these are a insignificantly small fraction of the group post-electronification.

Having established causality from financialization to market quality and pricing efficiency, we then carry out a structural vector auto-regression (SVAR) analysis to further examine the endogenous relationship between the participation of institutional financial traders and market quality. Using information from the same trader-level regulatory dataset from April 2007 to May 2008, we show that increases in the participation of institutional financial traders contemporaneously narrow bid-ask spreads, reduce customer trade imbalances, increase market depth, and cut the variance of pricing errors – all of which are unequivocally healthy developments.

The roadmap for the rest of this paper is as follows. Section II outlines our motivation and hypotheses, and places them in the context of extant literature. Section III describes our rich proprietary trading dataset from the regulator. We also define the specific measures of intraday pricing efficiency and liquidity we investigate. In Section IV, we document the huge beneficial impact of electronification on measures of market quality, and we provide also evidence of the financialization subsequent to electronification. Specifically, we show the large increase in the trading activity of institutional financial traders, thanks to the entry of new players; and we highlight that the increase in institutional financial trader activity is focused in the short-term futures contracts rather than the less liquid long-term contracts. Section V provides our rigorous analyses of the impact of financialization on market quality. In particular, it outlines the two-stage procedure we use to identify the causal link from financialization to market quality and pricing efficiency; and presents a structural VAR analysis linking changes in institutional activity and changes in various measures of quality. Section VI concludes and outlines avenues for further research.

II. Research Questions and Extant Literature

A. Intraday Pricing Efficiency

Our first research question is whether financialization improves intraday pricing efficiency, and in particular, the variance of “pricing errors”, i.e., the deviations of market prices from “fundamental” values, where the notion is based not just on the speed with which information is incorporated (correctly) into market prices, but also on the accuracy of such price adjustment. In this context, Sockin and Xiong (2015) argue that, because of globalization, today’s commodity market participants face severely heightened informational frictions (in relation to supply, demand, and inventories); and Singleton (2014) argues that these informational frictions, and the associated informational noise and speculative activity, can cause the magnitude of these pricing errors to drift upwards – increase – for extended periods. However, the focus of both of these papers is not on pricing errors over intraday horizons, but on relatively longer term pricing efficiency.

On the other hand, extant equity market based evidence shows strongly that, consistent with the predictions of Freidman (1953) and Fama (1965), institutions are better informed than other market participants, and that institutional trading increases information-based trading, and potentially reduces pricing errors. For example: (a) Boehmer and Kelley (2009) find a positive relation between institutional shareholdings and the relative informational efficiency of prices, measured as deviations from a random walk; (b) Chordia, Roll, and Subrahmanyam (2011) find that institutional trading results in an overall increase in information based trading; (c) Kumar, Thirumalai, and Yadav (2015) show that the informativeness of the trades of financial institutional traders is significantly greater than that of other traders; (d) Bartov, Radhakrishnan, and Krinsky (2000) find that institutional trading mitigates post-earnings drift; (e) Campbell, Ramadorai, and Schwartz (2009) infer that institutions take positions to arbitrage mispricings around earnings announcements; and (f) Badrinath, Kale, and Noe (1995) and Sias and Starks (1997) find that institutional trading reduces information processing

time.³ Particularly in the context of crude oil futures markets and the globalization argument of Sockin and Xiong (2015) cited earlier, this extensive evidence is consistent with the expectation that the increased presence of *institutional* financial traders in intraday futures trading and liquidity provision should arguably reduce informational imperfections since it brings greater access to global information and analytical resources relative to individual traders – the “Locals” who dominated futures markets prior to electronification.

Furthermore, financialization of intraday trading should also significantly increase the capital available with financial traders to fulfil more continually the risk sharing role of futures markets, i.e., the capacity to absorb the other side of hedging-related positions taken by commercial traders: and this should also reduce the variance of pricing errors by making risk sharing more efficient.

Overall, in view of the above, our first hypothesis is *H1: Financialization has a beneficial impact on the variance of intraday pricing errors.*

B. Intraday Liquidity

Our second research question is whether financialization improves intraday liquidity as manifested in bid-ask spreads, the Amihud depth measure, and the absolute value of trade imbalances. The role of institutional traders in liquidity provision is of increasing relevance for different financial markets, and to the best of our knowledge, this is the first paper that sheds light on this important topic.

There are several reasons why our expectation is that the impact of financialization on intraday liquidity should be beneficial. First, the increased presence of *institutional* financial traders in intraday

³ For other examples see, e.g., Grinblatt and Titman (1989, 1993), Daniel *et al.* (1997), Nofsinger and Sias (1999), and Bennett, Sias, and Starks (2003).

futures trading and liquidity provision should significantly increase the access to and availability of capital for liquidity provision, given that institutional liquidity suppliers arguably have much deeper pockets relative to individual traders. Second, institutional financial traders have access to better models for estimating short-term price changes based on information and liquidity flows, and greater ability to effectively manage their inventories and control risks; and, with the consequent reduction in “ambiguity” and costs, they can hence take greater position risks in individual liquidity provision trades. Third, the heavy increase in institutional financial trading necessarily increases competition among liquidity providers, potentially leading to more aggressive pricing. Each of the above three factors should arguably reduce spreads, increase depth, and reduce the imbalances between customer buys and customer sells.

In addition to the above, we have argued in sub-section A above that institutional traders are better informed than other market participants, and hence financialization – increased financial institutional traders – should increase information-based trading. We believe that this should, by itself, also increase intraday liquidity in the market. The theoretical models of Boulatov and George (2008, 2014) and Goettler, Parlour, and Rajan (2009) show that informed agents gravitate towards supplying liquidity rather than taking liquidity, and that is also consistent with the earlier empirical results of Kannel and Liu (2006). When the (more informed) institutional financial traders supply liquidity, they can do so arguably at lower cost since they need to make a relatively lower provision for adverse selection losses to more informed traders (Glosten and Harris, 1988). This competitive advantage in liquidity provision should also mean that the presence of (the relatively more informed) institutional financial traders in the market should, at the margin, lead to greater depth and lower trade imbalances.

Overall, in view of the above, our second set of hypotheses are:

H2A: Financialization reduces bid-ask spreads.

H2B: Financialization increases market depth.

H2C: Financialization reduces the absolute magnitude of customer trade imbalances.

C. Extant Literature

Our findings contribute to three strands of the extant financial economics literature. The first strand is the fast-growing literature on the financialization of commodity markets reviewed, for example, in Cheng and Xiong (2014).⁴ Prior evidence on the financialization of commodity markets is based almost entirely on daily, weekly, monthly, or even quarterly data on the ownership of overnight futures positions. In the case of crude oil markets, Büyükşahin *et al.* (2011) use non-public CFTC data on end-of-day futures and options positions to document strong growth in the overnight ownership share of hedge funds, commodity index, and other financial institutional traders in 2000-2010. Büyükşahin and Robe (2014) provide evidence of similar changes in position ownership patterns for a broad sample of commodities during the same decade, with a marked acceleration in 2004-2006. This paper adds significantly to the literature because, unlike extant articles, our focus is on the impact of the intraday positions and the intraday trading activity of institutional financial traders – essentially their contribution to intraday liquidity provision – to investigate the effects of financialization on *intraday* market quality.

The second strand is the large literature on the impact of institutional trading on financial markets. The literature relating to the impact on pricing efficiency is summarized in Section 2 subsection A above; and in that context, our empirical analyses and results complement earlier results,

⁴ In addition to the articles cited in the introduction, examples of research in this area include Brunetti and Reiffen (2014), Bruno, Büyükşahin, and Robe (2016), Büyükşahin and Harris (2011), Büyükşahin, Haigh, Harris, Overdahl & Robe (2011), Etula (2013), Hamilton (2009), Hamilton and Wu (2014), Hong and Yogo (2012), Irwin and Sanders (2012), Kilian and Murphy (2014), Knittel and Pindyck (2016), Korniotis (2009), Sanders, Irwin, and Merrin (2010), Stoll and Whaley (2010), and others.

especially those of Boehmer and Kelley (2009), by showing that institutional trading reduces the variance of commodity futures pricing errors. Our paper is among the first papers to address and investigate the impact on liquidity provision and price formation of short-horizon institutional traders with intraday holding periods: traders without any intrinsic motivation or pre-commitment to trade.⁵

Finally, our intraday focus on financialization connects our work to prior market microstructure research on the impact in financial markets of structural changes in trading activity reviewed, for example, in Holden, Jacobsen, and Surahmanyam (2014), and Chordia, Roll, and Subrahmanian (2011). The closest paper to ours in that market microstructure strand of literature is Hendershott, Jones and Menkveld (2011), who analyze the impact of algorithmic trading (AT) on market quality using the rate of electronic message traffic as a proxy for AT. They use the staggered introduction of autoquote by the NYSE in 2003, and the subsequent surge in electronic messaging for autoquote-enabled stocks, to ascertain whether AT improves market liquidity. Our underlying empirical approach is similar, but with a key difference. Econometrically, we also use an exogenous event, i.e., electronification, as an instrument – in our case, to extract the component of institutional financial trading that is exogenous to market conditions – but our objective is to identify the impact on market quality of financialization – not of electronification. To delineate financialization, we exploit economically and statistically significant differences in institutional financial traders’ shares of the total trading activity in near- vs. far-dated crude oil futures before and after electronification.⁶

⁵ Raman and Yadav (2015) and Kumar, Thirumalai, and Yadav (2015) examine, *inter alia*, the role of individual and different types of institutional traders on liquidity provision and pricing efficiency respectively. XX Others?

⁶ Hendershott *et al.*'s (2011, p. 16) “conclusions on causality rely on the intuitively appealing but ultimately untestable assumption that autoquote affects liquidity only via its effect on AT.” In a similar vein, a possible concern in our setting could be that changes in the behavior of trader types other than institutional financial traders might be responsible for the market quality improvements we observe. We rule out this alternative by showing that changes in the trading activity of other groups of traders cannot be credited for the improvements.

III. Data & Measures of Market Quality

A. Data

We use intraday data on all futures transactions in U.S. crude oil futures, and for every trading account over a specific period. This information is non-public and originates from the U.S. Commodity Futures Trading Commission (CFTC), the market regulator which collects this information. The New York Mercantile Exchange (NYMEX), acquired by the Chicago Mercantile Exchange (CME) group in 2008, introduced electronic trading in September 2006 for four energy futures markets, and in December 2006 for three metal futures markets. For this paper we were given access to non-public, trade-level regulatory data for the largest of those seven markets, i.e., for West Texas Intermediate (WTI) crude oil futures.

We aggregate the account-level data across multiple contracts and into three trader categories – Locals (CTI-1), institutional financial traders (CTI-2), and customers (CTI-4).⁷ These three trader categories comprise dozens, hundreds, or thousands of trading accounts, in a manner that preserves trader anonymity and trading secrets.⁸ Each futures trade is recorded twice, once by the buyer and once by the seller. The buyer and the seller are each identified only by an identity code in the dataset. Those anonymizing codes are assigned by the CFTC to each trading account so as to conceal the actual identities of market participants. Hence, while the data to which we had access provides a complete WTI trading history for each trader for a specific period, each trader's true identity remains confidential and unknown to us.

⁷ Only about four percent of all trades are classified as CTI-3 (individual member trading on behalf of another member). These trades are largely not relevant for this paper.

⁸ Consistent with the confidentiality statutes under which the CFTC operates, no single or multiple set of information that we provide in any part of the paper could allow anyone to uniquely identify any trader's underlying position(s) or trade secrets and strategies.

The data made available by the CFTC for the purposes of our study include the trade records for all transactions executed over the 28-month period between January 3rd, 2006 and May 31st, 2008 for all the WTI sweet crude oil futures traded at NYMEX. The CFTC dataset includes transaction details such as the commodity and delivery month, the quantity traded, the trade type (outrights, spreads, trades at settlement, etc.), the trade's price and direction (i.e., whether the transaction was a buy or sell), and the transaction's date and time. For electronic trades on the CME Globex platform, the latter is the time stamp assigned to the trades when both sides were matched. For open outcry trades that were done in the pits, it is the imputed trade time stamp. For our analysis, we use (*pre-electronification*) Pit trades that are time-stamped between 10AM-2:30PM and (*post-electronification*) Globex trades that are time-stamped between 9AM-2:30PM.⁹

B. Measures of Market Quality

We employ several measures to assess market quality: (a) the volatility of pricing errors, i.e., the average magnitude of deviations of prices from their “fundamental” values; (b) the bid-ask spread; (c) the Amihud measure of depth; and (d) customer order (trade) imbalances. We compute all variables for each contract maturity on every trading day in five-minute non-overlapping intervals, then we compute daily volume-weighted averages of the five-minute figures during business hours.¹⁰ On any given day in our sample period, futures with up to 84 different maturities are traded.

⁹ Pits used to be open from 10AM-2:30PM prior to January 31, 2007; starting February 1, 2007, pit business hours were increased to 9AM-10:30PM. We exclude from the sample the Friday immediately after Thanksgiving as well as the entire week from Christmas to New Year (starting the last full trading day before Christmas and ending the first trading day after New Year). Before aggregating the data, we carry out a number of quality checks; for example, we exclude transactions whose reported prices are clearly erroneous.

¹⁰ Trading hours are 9AM to 2:30PM for electronic contracts starting September 5th, 2006. Pit trading hours are 10AM to 2:30PM until January 31st, 2007 and 9AM-2:30PM thereafter.

We estimate the volatility of pricing error as in Hasbrouck (1993). The logarithm of the observed transaction price, p_t , is expressed as the sum of the logarithm of the efficient price, m_t , and the pricing error, s_t as follows:

$$p_t = m_t + s_t$$

The pricing error is a measure of how efficiently the transaction price tracks the (unobserved) fundamental price, represented by an information efficient “random walk price”, and is an excellent measure for testing pricing efficiency (Fotak, Raman, and Yadav, 2014). Since the pricing error is a zero-mean process, its absolute magnitude is a good measure of its volatility. We follow Hasbrouck (1993) and Boehmer and Kelly (2009) and estimate the lower bound of the volatility of the pricing error, σ_s , using a VAR system consisting of four variables: $\Delta p_{t,i}$, trade sign indicator (estimated using tick-test), signed trading volume, and the signed square root of the trading volume. We compute σ_s on every trading day for each contract maturity for which at least 50 trades take place. In our tables, the variable “*PE_Proportion*” is the daily ratio of the variance of pricing errors, estimated as in Hasbrouck (1993), to the volatility of intraday (log) transaction prices.

We estimate daily bid-ask spreads to approximate the cost of transacting for customers of the exchange. In the absence of order-book information, we estimate bid and asked prices for each contract maturity in each contract interval, after classifying trades as buyer- vs. seller-initiated using Lee and Ready’s (1991) “tick-test.”¹¹ In our tables, the variable “*Spread*” is the daily average of 5-minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each 5-minute interval, after classifying trades as buyer- vs. seller-initiated using the Lee and Ready (1991) tick-test. We calculate the inverse measure of depth as proposed by Amihud (2002), reported

¹¹ Raman, Robe, and Yadav (2014) verify the accuracy of the tick-test using data from a time period (post-2009) when the “aggressive” trader starts being identified by a flag in the CFTC’s non-public intraday dataset. In the case of WTI futures, these authors find that the tick-test successfully identifies the (actual) aggressive trader in more than 75% of the cases – similar to the 73% figure found by Theissen (2001) using Frankfurt Stock Exchange data.

in our tables as the variable “*Amihud*”, as the average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. We calculate daily volatility as the average of the maximum of customer (i.e., CTI-4) buy price and sell price volatility, calculated in five-minute non-overlapping intervals throughout the day. This methodology is meant to eliminate the bid-ask bounce (Manaster and Mann, 1996). Finally, we measure daily customer trade imbalances, reported in our tables as the variable “*AbsOIB*”, as the average of the ratio of five-minute absolute trade imbalances (buyer-initiated *minus* seller-initiated trades) to trading volume.

IV. The Impact of Electronification

A. Market Quality Metrics and Overall Financial Institutional Trader Activity

NYMEX introduced electronic trading on Globex on September 5, 2006 in parallel with face-to-face trading in the pits intermediated largely by “Locals”. In this sub-section, we provide a descriptive analysis of the change in market quality measures around electronification, and also of changes in the trading of institutional financial traders over the same period. Given that this market reform significantly increased ease of access for traders without physical access to the trading pits, and correspondingly increased competition in liquidity provision services, one should expect a marked improvement in each of our market quality measures. The results presented in Table 1, and displayed in Figure 1 Panels A to D show that this is indeed the case. We should also expect an increase in the proportion of institutional financial traders after electronification. This is exactly what we do find. The results are in Table 2, and displayed in Figure 2.

Table 1 and Figure 1 Panels A to D, present a descriptive analysis of our measures of market quality surrounding the introduction of electronic trading on September 5, 2006. Table 2 and Figure 2 provide a corresponding descriptive analysis of the nature and extent of participation of Locals,

institutional financial traders, and customers. The sample period for Table 1 and Table 2, and for Figure 1 Panels A to D and Figure 2, is the period from January 3, 2006 to March 31, 2007. *Pre-Electronification* in the Table refers to the period over January 3, 2006 to September 1, 2006 and *Post-Electronification* refers to the period from September 5, 2006 to March 31, 2007. The figures relate to WTI sweet crude-oil futures Pit trading in the *Pre-Electronification* period, and to the Globex platform in the *Post-Electronification* period. The market quality measures in the table have been defined earlier in Section 3 sub-section B. *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI-2 traders in the CFTC database), and is hence our measure of financialization.

Table 1 shows a staggeringly large improvement in market quality measures following electronification. The corresponding Figures – Figure 1 Panels A to D – provide strong visual confirmation. Spreads drop from an average of 37 basis points to just 3 basis points, a drop of more than 90%. Absolute order imbalances drop about 40% from about 24% to about 13%. The ratio of the variance of pricing errors to the variance of log transaction prices changes from about 59% to about 4%, again a drop of more than 90%. And each of these changes are statistically highly significant, with a p-value $\ll 0.001$. The only measure which does not show a statistically significant change is the Amihud measure.

Table 2 shows a concomitant large change in financialization accompanying electronification. The average proportion of trading activity involving institutional financial traders almost doubles from 29.6% pre-electronification to 55.0% post-electronification. Figure 2 provides visual trading activity-based evidence of commodity markets' financialization by aggregating the CFTC's account-level intraday information by trader type. Prior to electronification, consistent with Manaster and Mann (1996), Locals dominated Pit trading: in the first eight months of 2006, Locals were involved

on at least one side of approximately 80% of all WTI futures transactions. In contrast, CTI-2 traders were involved (on at least one side of the transaction) in only about 30% of all Pit trades during the same period. However, following the electronification of the WTI futures market, Figure 2 shows that the proportion of CTI-2 trades on Globex approximately doubles, and the proportion of trades with Locals reduces to approximately half of its pre-electronification value. Over the same period, the overall volume increased substantially as well. Hence, in a market that Figure 2 shows to have grown quickly, the slice of financial institutional traders in the growing WTI futures pie also grew markedly. Finally, and very interestingly, Table 2 shows that the median value of the absolute closing ratio – the average ratio of a trader’s ending-of-hour inventory and hourly trading volume – declined sharply from 83% prior to electronification to only 8% after electronification, indicating that the newly entering institutional financial traders were focused on much shorter horizon strategies – similar to the strategies of Locals – relative to the institutional financial traders that had been active earlier.

These results clearly show that the introduction of electronic trading dramatically improved market quality in the crude-oil market, as it earlier did in the equity markets (Jain, 2005); and in the case of the crude oil market, it was accompanied by financialization focused on short-horizon intraday institutional financial traders. Clearly, the above descriptive analysis cannot differentiate the respective impacts of electronification and financialization. Since financialization is the primary focus of this paper, we next focus on differences in the participation of institutional financial traders between futures contracts of different maturities.

B. Institutional Financial Trading: Short-term vs. Long-term Futures Contracts

With a view to distinguishing between the impact of electronification and financialization, we note that electronic trading can improve market quality through three main channels. Firstly, it

improves *pre-* and *post-*trade transparency in the market, and thereby reduces information asymmetry (Boehmer, Saar and Yu, 2005). Secondly, it reduces fixed labor (pit traders), and other operating and order processing costs. Finally, it drastically cuts the costs of entry into provision of liquidity services, and enables all exchange members, irrespective of their physical location, to exploit small deviations from fundamental value and provide liquidity, thereby significantly increasing competition: the open and transparent electronic market, where all traders have an equal opportunity to voluntarily provide and demand liquidity, attracts new groups of traders, particularly institutional financial traders (Jain, 2005). Of these three channels, the first and the second should equally impact futures contracts of all maturities, and both short- and long-term contracts should equally benefit from improvements in transparency and from the reduction of fixed ordering costs.

However, all contract maturities are not expected to experience the same amount of interest from institutional financial traders. In particular, institutional financial traders have shorter trading horizons than other traders. Hence, they should arguably trade more in short-term than in long-term contracts (Ederington and Lee, 2002). Hence, with electronification attracting new financial traders, the trader-composition of the WTI sweet crude oil futures market can potentially change differentially for short-term and long-term contracts. We find empirically that this is indeed the case.

We compute participation rates of institutional financial traders and our market quality metrics separately for two groups of futures contracts: short-term contracts *vs.* long-term contracts. Given that approximately half of the WTI futures trading volume involves spread trades in our sample period, our “short-term contracts” bin consists of the trading activity in the two front months (contracts with less than 62 days to expiration), *and* the trading activity in the four December contracts with which these two nearest-dated contracts are often paired with for calendar spreads.¹² Our “long-term

¹² In our sample period, crack spread trades accounted for about 3.7% of all transactions and 1.8% of the trading volume in WTI futures. Calendar spreads accounted for 22.2% of all WTI futures trades and 50.1% of the total trading volume.

contracts” bin consists of trading activity in all remaining contracts with more than 62 days to expiration.

Figure 3, Panel A plots the evolution of institutional financial traders’ participation in short-term and long-term contracts. The figure shows a much stronger rate of financialization for short-dated contracts relative to long-dated contracts. Also, Figure 3, Panel B plots the number of new financial institutional traders entering the crude oil futures market in short-term and long-term contracts. Consistent with the evolution of participation rates, we find that there are a significantly greater number of new institutional financial traders entering the trading of short-term contracts than of long-term contracts. Overall, the bottom line is that the introduction of electronic trading encouraged an influx of new financial institutional traders, albeit largely in the short-term contracts. Table 3 presents a descriptive analysis of, and a more formal comparison, of the differences pre- and post-electronification in the proportions of institutional financial traders – i.e., the extent of financialization – between short-term and long-term futures contracts. While the proportion of financial trading in short-term contracts – $FIN_Short-Term$ – increases from 28.1% to 41.4% (an increase of almost 45%, statistically significant with p-value less than 0.001), the equivalent measure for long-term contracts – $FIN_Long-Term$ – does not change significantly in long-term contracts, remaining around 36%-37% both before and after electronification. As consequence, the percentage/proportional difference in the fraction of financial trading ($\Delta FIN: (FIN_Short-Term - FIN_Long-Term) / FIN_Short-Term$) increases from -30.4% to 9.5% around the electronification of crude-oil markets. The event-related increase is statistically significant at the 1% level.

Clearly, the participation of financial traders (financialization) in short-term and long-term contracts changes differently due to electronification, an exogenous exchange-mandated intervention:

it is much greater for short-term contracts relative to long-term contracts.¹³ In the next section, we exploit the differences in the extent of financialization between short-term and long-term futures contracts to formally test, in an econometrically rigorous framework, for the impact of financialization on market quality. Essentially, even though the market quality metrics for all contracts improve post-electronification, what we test for is whether the extent of market quality improvement post-electronification is greater for contracts with a greater entry of new institutional financial traders, after controlling for any other factors that could be relevant.

V. Financialization and Market Quality

We have seen in Section IV that our market quality metrics improve overall significantly post-electronification. We have also seen that financialization in short-term contracts in response to this electronification is much greater than that in long-term contracts on the same asset. In this Section, we accordingly examine the causal effect of financialization on market quality by exploiting this observed cross-sectional variation in financialization across different contract maturities after the (exogenous) exchange-mandated electronification of the crude-oil futures market. Our contention is that the *average* improvement in market quality variables across all contracts is the effect of electronification, but the *relative* improvement we find for the short-term contracts is due to the relative financialization of the short-term contracts.

Section V.A presents a descriptive analysis of the difference, pre- and post-electronification, in our market quality metrics for short-dated and long-dated futures contracts. This is a simple event study conducted around the introduction of electronic trading. Section V.B presents a formal two-

¹³ The significant increase in the proportion of financial institutional trading after the onset of electronic trading echoes the massive growth of the overnight WTI futures positions held by hedge funds and other financial traders during the same period (Büyüksahin *et al.*, 2011).

stage regression analysis. In Section V.C, we show how the the results remain unaffected when we do not include high speed traders, and are hence not being driven by algorithmic or high speed traders. Then, having established causality through the two stage regresssion analysis, we use in Section V.D., a structural VAR (SVAR) analysis to further examine the endogenous relationship between participation of institutional financial traders institutions and market quality using data between April, 2007 and March, 2008, a period of stable electronic trading.

A. Descriptive Analysis – Event Study

In this sub-section, we present results of an event study conducted to a sample comprising all the transaction records for WTI futures on NYMEX covering and eight-month period before and a seven-month period after the onset of electronic trading, i.e., all trading days between January 3rd, 2006 and March 31st, 2007. We employ the market quality measures discussed in the previous section, and we compute all daily averages separately for two groups of contract maturities – short-term and long-term contracts. Table 4 presents the pre- and post-electronification values of our market quality variables, and the differences, for short-term and long-term contracts separately. The results are also graphically presented in Figure 4. Panels A to D of Figure 4 plot the evolution of the percentage differences (trade-volume weighted for short-term vs. long-term contracts) in each of the market quality measures. Figure 4 shows that, for each of the measures, the improvement after electronification benefited short-term contracts more than the long-term contracts.

As is clear from Table 4, (a) the pre-electronification average ratio of the volatility of pricing errors to the volatility of log transaction prices is 28% wider than that of short-term contracts, but 132% wider in the post-electronification period, increases almost five times: while both long- and short-term contracts have much lower pricing errors post-electronification, the pricing errors of short-

term contracts improve significantly more than that of long-term contracts (p-value $\ll$ 0.001); (b) the pre-electronification average bid-ask spread of long-term contracts is about 28% wider than those of the short-term contracts, and the same variable in the post-electronification period increases more than ten times in magnitude to 304%: while both long- and short-term contracts are more liquid post-electronification, the liquidity of short-term contracts improves significantly more than that of long-term contracts (p-value $\ll$ 0.001); (c) the pre-electronification average magnitude of customer order imbalances of long-term contracts is about 144% wider than those of the short-term contracts, and the same variable in the post-electronification period increases significantly to 175%: while both long- and short-term contracts display lower customer order imbalances post-electronification, the improvement for short-term contracts is significantly more than that of long-term contracts (p-value $\ll$ 0.001); and finally (d) the pre-electronification average depth (inversely measured by the Amihud ratio) is about XXX% wider than those of the short-term contracts, and the same variable in the post-electronification period increases significantly to XXX%: while both long- and short-term contracts display higher depth post-electronification, the improvement for short-term contracts is significantly more than that of long-term contracts (p-value $\ll$ 0.001). Meanwhile, as discussed in the preceding Section, the participation of institutional financial traders' volume increases significantly in short-term contracts while remaining (statistically) the same in the long-term contracts.

Hence, for each of these market quality metrics, electronification benefits short-term contracts significantly more than it does long-term contracts, and the results are consistent with our contention that the *average* improvement across all contracts is the effect of electronification, but the *relative* improvement of the short-term contracts is due to the relative financialization of the short-term contracts. Prima-facie, notwithstanding the lack of relevant controls like changes in relative trading volume and volatility, it appears that financialization improves market quality over and above the

improvement that comes from electronification. The analyses in the following sub-sections examine this conjecture more rigorously in a multivariate setting.

B. Two-Stage Regression Analysis

The introduction of electronic trading removed barriers to participation in the crude oil futures market and facilitated market participation by financial institutional traders. We have shown earlier that there has been a huge increase in financial institutional trader activity post-electronification. Under the assumption that the NYMEX's decision to "go electronic" was exogenous with respect to pre-existing crude oil derivatives market conditions, we use the advent of WTI futures market electronification as an instrument to tackle the endogeneity issues between market quality and trading activity of financial institutional traders. As discussed earlier, our goal is to identify the impact on market quality of financialization – not of electronification. To tease out the impact of financialization, we exploit the exogenous increase due to electronification in the *relative* participation of financial institutional traders in short-term (vs. long-term) contracts in order to examine the impact of financialization on different measures of market liquidity and pricing efficiency.

In particular, we use the electronification of the crude-oil futures markets as an instrument for the *relative* participation of financial institutional traders in short-term (vs. long-term) contracts in our analyses as follows:

$$\text{First Stage:} \quad \Delta FIN_t = \alpha_1 + \beta_1 \text{Electronification} + \gamma_1 C_t + \theta_t$$

$$\text{Second Stage:} \quad \Delta M_t = \alpha_2 + \beta_2 \widehat{\Delta FIN}_t + \gamma_2 X_t + \epsilon_t$$

where ΔFIN_t is the percentage (or proportional) difference between fraction of participation of institutional financial traders in short-term and long-term contracts', i.e., (ΔFIN : ($FIN_Short-Term -$

$FIN_Long-Term)/FIN_Short-Term)$; $\widehat{\Delta FIN}_t$ is the predicted value obtained from the first stage; ΔM_t is the relative difference between short-term and long-term contracts' relevant market quality measure (such as Bid-Ask spreads), *Electronification* is a dummy variable equal to 1 after September 5th, 2006 (the date on which electronic trading was introduced), and 0 prior to that day; and X_t and C_t are control variables. β_2 is our estimate of the (causal) effect of financialization on market quality.

In the first stage, we regress the relative difference between institutional financial traders' rates of participation in trades of short-term and long-term contracts on the *Electronification* dummy, and on several binary variables selected to control for possible differences in trading patterns on: (i) nearby expiration days; (ii) days when the U.S. Department of Energy's Energy Information Administration (EIA) releases its weekly reports on petroleum inventories, as well as the day prior to the news release; (iii) days when commodity index traders (CITs) that follow the GSCI indexing methodology roll their near-dated futures positions; (iv) and day-of-the-week trading patterns. We also control for market-wide uncertainty using equity option-implied volatility (the VIX).

The results are presented in Table 5. Consistent with the univariate and graphical evidence presented earlier, we find a statistically and economically significant relation between *Electronification* and the difference in institutional financial traders' participation in short-term vs. long-term contracts. The coefficient of 0.40 implies that, *post*-electronification, the relative participation of financial traders in short-term contracts increased by as much as 40 percentage points. Prior to electronification, institutional financial traders participated more in the long-term contracts than in the short-term contracts, as indicated by the intercept of 0.30. This results holds whether (second column in Table 5) or not (first column) we include relevant control variables in the regressions.

Importantly, we find that the *Electronification* dummy alone explains more than 30 percent of the variation in ΔFIN , indicating that it is a strong instrument for the change in the proportion of institutional financial traders. That the exogenous shock of electronification strongly predicts a significant increase in the financialization of short-term contracts relative to long-term contracts is a necessary condition for the second stage regressions.

In the second stage, we then use the predicted values for the percentage difference in institutional financial traders' participation in short-term and long-term contracts obtained from the first regression, $\widehat{\Delta FIN}$, as a dependent variable when regressing our various measures of market quality (precisely, the percentage difference in depth, spreads, order imbalances and pricing error volatility between short-term and long-term contracts) on trading volume and realized volatility (precisely, the percentage differences between short-term and long-term contracts) and on the dummy variables of the first stage (near-contract expiration day, GSCI roll-day dummy, days of or preceding EIA inventory announcements, and Mondays or Fridays). In line with the results of Breusch-Godfrey tests, we also include two lags of the dependent variables; we estimate standard errors using the Newey-West method with two lags. Our results from the second stage analysis are summarized in Table 6 for the variance of pricing errors, Table 7 for bid-ask spreads, Table 8 for depth, and Table 9 for customer order imbalances.

For each and every market quality metric – variance of pricing errors in Table 6, bid-ask spread in Table 7, depth in Table 8, and customer order imbalances in Table 9 – $\widehat{\Delta FIN}$ consistently has a negative and statistically highly significant coefficient (p-value $\ll 0.001$) in all specifications relating to that market quality variable. To interpret the result, recall that we are analyzing the relation between percentage differences in that particular market quality variable and percentage differences in financialization. In this context, our result means that an increase in the percentage (or proportional)

difference in participation of institutional financial traders in short- vs. long-term contracts causally influences the percentage (or proportional) difference in that market quality variable between short- vs. long-term contracts. That constitutes strong evidence that financialization improves each of these market quality variables. For the regressions for each of these market quality variables, we include the percentage difference in volatilities between short-term and long-term contracts, and we include the percentage difference in trading volumes as regressors. We also introduce a *September, 2006* dummy variable to control for the transitory irregularities during the month when electronification took place. Still, we find that even after controlling for the differences in volatility and volume, EIA information announcement days, contract roll and expiration days, when short-term contracts experience a greater degree of financialization than long-term contracts, the market quality variable for short-term contracts improves significantly more than the corresponding market quality variable of long-term contracts.

Let us consider each of the market quality variables separately. Table 6 shows that a one-unit increase in the percentage difference in financialization between short-term and long-term contracts, after controlling for all the relevant variables, further widens the percentage difference in pricing error volatility by 1.03 units in Model 1 (and by 0.44 units on average across all specifications). The standard deviation for the percentage difference in financialization is 36%; hence, a one-standard deviation increase in the percentage difference in the rate of financialization, widens the percentage differences in pricing error volatility by 0.35 standard deviations, or 59% of its mean value. Table 6 hence strongly shows that financialization significantly improves pricing efficiency. In our analysis of the volatility of pricing errors, we follow Boehmer and Kelly (2009) in including the standard deviation of transaction *prices* as control while modeling the volatility of the pricing error. Accordingly, we employ the percentage difference in the volatility of transaction *prices* between

short-term and long-term controls while modeling the percentage difference in pricing error volatilities.

For bid-ask spread, the coefficient of -4.22 in Model 1 in Table 7 means that a one-unit increase in the percentage difference in financialization (where the participation of financial traders in short-term contracts increases more than it does in long-term contracts) decreases the percentage differences in bid-ask spreads (where the spreads for short-term contracts decrease more than spreads for long-term contracts) by 4.22 units. The standard deviation for percentage difference in bid-ask spreads is 186%; so, a one-standard deviation increase in the percentage difference in financialization widens the percentage differences in bid-ask spreads by 0.76 standard deviations, or 94% of its mean value. Table 7 also shows that, as should be expected, the percentage difference in spreads is positively related to the percentage difference in volatilities between short-term and long-term contracts and negatively related to the percentage difference in trading volumes.

Our results from the analysis of the differences in depth (inversely measured by the Amihud ratio) are similarly presented in Table 8. Controlling for all the previously discussed variables, and similar to the bid-ask spread results, we find a consistently negative and statistically significant relation between the difference in the financialization rate and the difference in Amihud ratios. A one-standard deviation increase in the percentage difference in financialization widens the percentage difference in the Amihud ratios by 0.80 standard deviations, or 28% of its mean value.

Results from the analysis of the differences in customer trade imbalances are presented in Table 9. We again find a negative and statistically significant relation between the difference in financialization and the difference in customer trade imbalances. A one-standard deviation increase in the relative difference in the rate financialization, widens the relative differences in customer trade imbalances by 0.58 standard deviations, or 19% of its mean value.

To summarize, we find that when short-term contracts experience a greater degree of financialization than long-term contracts, the market quality of short-term contracts, measured in terms of bid-ask spreads, depth, customer trade imbalances or pricing error volatility, improves statistically and economically significantly more than the market quality of long-term contracts.¹⁴

C. Are the Results Driven by High Speed Traders?

Hendershott, Jones, and Menkveld (2011) show that algorithmic trading improves intraday market quality metrics similar to our conclusion that financialization improves these metrics. It could be argued that our results are driven by the rise of algorithmic or high speed traders. To rule out this possible confounding explanation, we repeat the analysis after removing all “fast” liquidity providers (FLPs) from our sample of institutional financial traders. We identify FLPs as traders who make more than 1,000 trades a day and carry less than 5% of their daily trading volume overnight (Raman, Robe and Yadav, 2014). Results from the second-stage regression for non-fast CTI-2 traders are presented in Table 10. They are qualitatively similar to those reported in Tables 6 to 9 for all institutional financial traders. For each market quality metric – variance of pricing errors, bid-ask spread, depth, and customer order imbalances – $\widehat{\Delta FIN}$ for just the “non-fast” institutional financial traders has a negative and statistically highly significant coefficient (p-value<<0.001). Hence, it is clear that our results are not driven by the arrival of algorithmic or high speed traders.

¹⁴ The two-stage analysis in this sub-section focuses on CTI-2s, to the exclusion of CTI-1s. One might expect that a two-stage regression analysis using the volume share of CTI-1s (rather than that of CTI-2s) might produce “mirror” results. In that case, it could be argued, albeit facetiously, that it is the crowding out of CTI-1 traders, and of their practices in the trading Pits (rather than the competition from institutional financial traders) that led to the *post*-electronification increase in market quality? However, this possibility is unlikely since CTI-1 trading volume (as opposed to the CTI-1 *share* of the total volume) actually *increased* following electronification – thus suggesting that it is indeed greater competition from CTI-2s, rather than the disappearance of CTI-1s, that drove the improvement in market quality.

D. VAR Analysis

We further examine the endogenous relation between participation by financial institutions and market quality through structural VARs. We rely on the same comprehensive dataset of intraday transactions, but restrict the sample period from April 1st, 2007 to May 31st, 2008 in order to avoid any overlap with the two-stage regression sample period and to test whether our results from the previous section persist even in stabilized market conditions.

As in the previous tests, we use the percentage differences in financialization, bid-ask spreads, Amihud measure, absolute customer imbalances, and pricing error volatility. We also control for the percentage difference in volatility. Augmented Dickey-Fuller (ADF) tests show that all of the percentage difference variables used in the SVAR analyses are stationary. The number of lags is chosen according to the Akaike information criteria. As the focus of this paper is on the effect of financialization on other variables, we place ΔFIN first in the ordering of the SVAR variables. The results are robust to different orderings of the remaining variables. ...

The impulse response functions, showing the effect of a one-standard deviation shock to ΔFIN on $\Delta Spread$, $\Delta Amihud Measure$, ΔABS_{OIB} and $\Delta PE_Volatility$ are presented in Panels A to D of Figure 5. Consistent with the results of the event-study and of the regression analyses, Panel A shows that an increase in ΔFIN results in a negative and significant effect on (i.e., an improvement in) the relative difference in bid-ask spreads both on day t (contemporaneously) and on day $t+1$. An increase in the relative financialization of short-term contracts leads to a decrease in their relative bid-ask spreads. While the effect of ΔFIN on $\Delta Amihud$ is insignificant (Panel B), we find a negative and significant effect on the change in absolute order imbalances, i.e., ΔABS_{OIB} (Panel C); and on the change in pricing errors, $\Delta PE_Volatility$ (Panel D). More specifically, a one-standard deviation increase in ΔFIN leads to a 0.19 and 0.15 standard deviations

decrease in contemporaneous ΔABS_OIB and $\Delta PE_Volatility$ respectively; and a further 0.13 standard deviation decrease in $\Delta PE_Volatility$ on day $t+1$.

V. Conclusion

The NYMEX introduced electronic trading to its energy futures marketplace on September 5th, 2006. Focusing on West Texas Intermediate (WTI) sweet crude oil futures, we document that this change in market structure increased the participation rate of institutional financial traders – i.e., financialization – while leading to a relative drop in the activities of Locals – the traditional liquidity providers in these markets. We use this momentous event as an exogenous instrument to measure the causal effect of financialization on WTI market quality (magnitude of pricing errors, bid-ask spreads, depth, and order imbalances). Exploiting cross-sectional variation in the rates of financialization, we document economically and statistically significant improvements in all these market quality proxies due to financialization. *Post-event*, we then use structural VAR analyses to further examine the endogenous relationship between participation by financial institutions and market quality. This SVAR analysis provides additional strong evidence that greater participation by institutional financial traders brings about a significant improvement in bid-ask spreads, absolute order imbalances, and the volatility of pricing errors.

Our results point to several venues for future research. First, liquidity provision in U.S. commodity futures markets is entirely voluntary, and given the significant increase we document in the extent and influence of institutional financial trading, an important question from a regulatory perspective is whether, in periods of stress, institutional financial traders make markets more or less resilient in the manner of Kempf, Mayston, Gehde-Trapp, and Yadav (2016) – or more or less fragile in the sense of Raman, Robe, and Yadav (2014). One way of potentially addressing this question by

exploiting differences in the extent to which institutional financial traders have come to dominate liquidity provision in commodity futures *vs.* options on futures markets. That is, one could exploit this difference in order to investigate – by way of a “difference-in-differences” analysis – whether the financialization of commodity markets has an effect on their resilience to (in particular, on liquidity provision after) an exogenous shock. Those findings would have implications for financial stability and the importance of systemic risk in the presence of electronic trading.

Finally, the analyses that we have run has focused on intraday data – without separating the contribution of traders who provide short-term liquidity from that of longer-term liquidity providers (who might carry inventories overnight). Non-commercial WTI overnight positions, however, increased substantially (by one third to one half, depending on the category of futures trader) right after the onset of electronic trading. Further research might therefore also consider the contribution of longer-term liquidity providers to market quality. Of particular interest would be the potential impact of electronification on commodity risk premia. Intuitively, one could expect an impact if, by reducing the profitability of liquidity provision by locals, electronification affects the provision of longer-term liquidity. We leave this, and other questions, for further research.

VI. References

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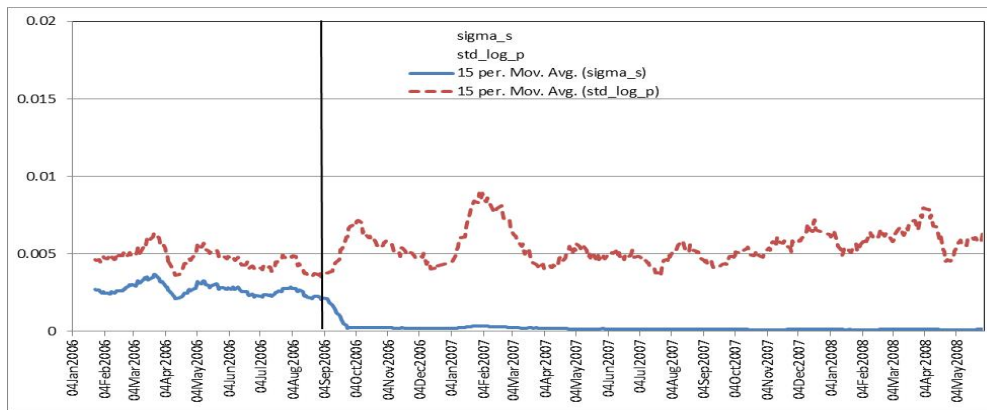
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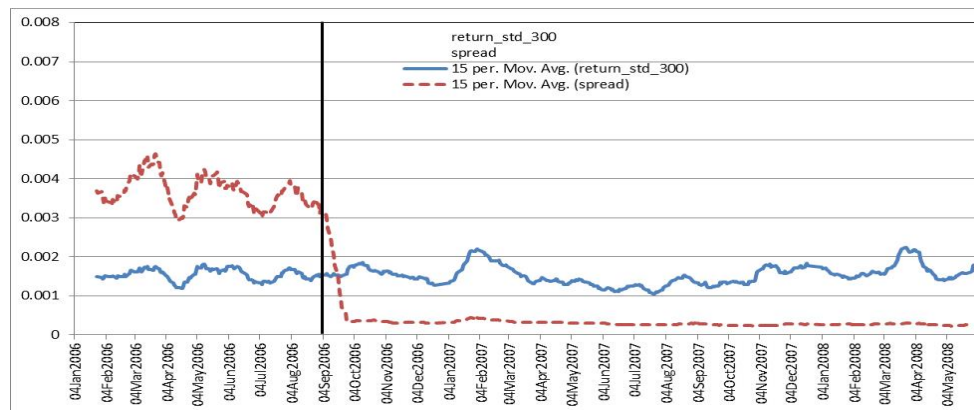
Figure 1: Market Quality Measures

Figure 4 depicts the evolution of various market quality measures in the NYMEX WTI sweet crude oil futures market between 2006 and 2008. Measures are computed in the Pits for the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to May 31st, 2008). *Spread* is the daily average of 5 minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each interval (5 minutes), after classifying trades as buyer- vs. seller-initiated using the Lee and Ready (1991) tick test. *Amihud*, an inverse measure of depth (Amihud, 2002), is the daily average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. *AbsOIB* is the daily average of 5-minute customer trade imbalances (traders classified as CTI-4 traders in the CFTC database) calculated as the ratio of five-minute absolute trade imbalances (buyer-initiated minus seller-initiated trades) to trading volume. *PE_Variance* is the daily pricing error variance, estimated as in Hasbrouck (1993). *Price_Volatility* is the daily average of 5-minute volatility of intraday (log) transaction for each contract maturity in each interval. All the measures are estimated for each contract maturity and then volume-weighted averages across all 84 futures contract maturities are plotted.

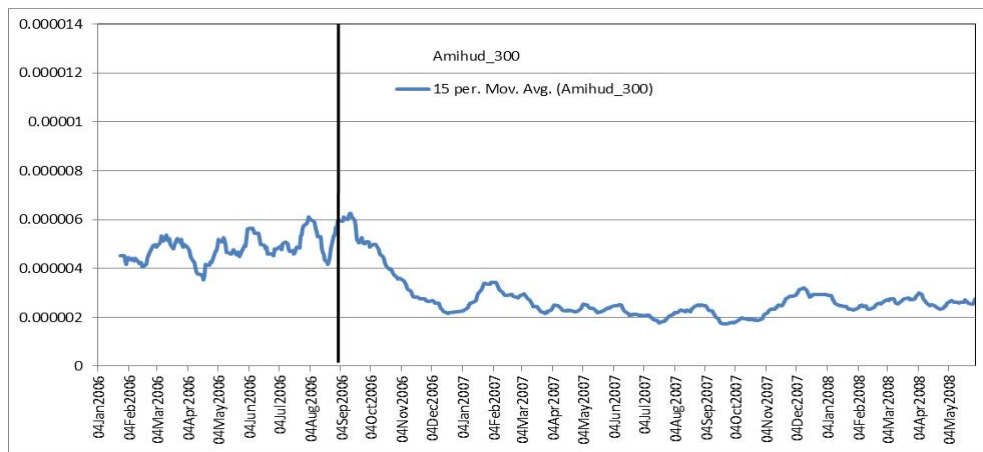
Panel A: Pricing Errors (*PE_Variance* and *Price_Volatility*)



Panel B: *Spread* and *Volatility*



Panel C: Inverse Depth (*Amihud Ratio*)



Panel D: Customer Demand Imbalances (*AbsOIB*)

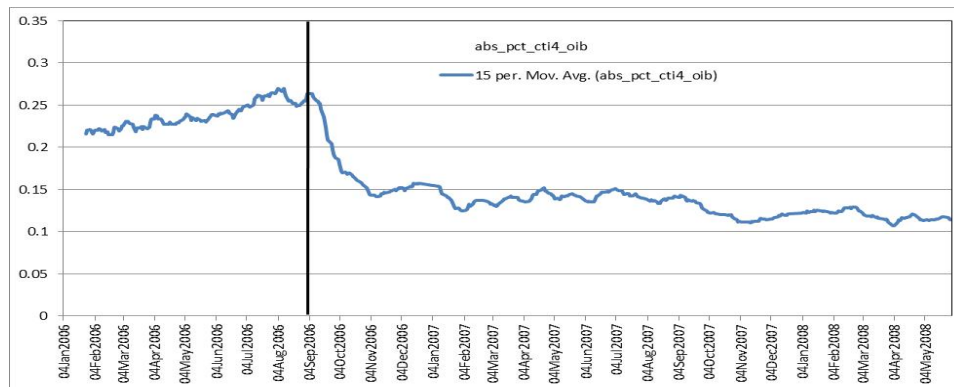


Figure 2: Trading Volume in Short-Term and Long-Term contracts

Figure 2 compare the evolution of the proportion of trading volume involving financial traders and Locals in the NYMEX's WTI sweet crude oil futures market between 2006 and 2008. Measures are computed in the Pits for the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to May 31st, 2008). *FIN* is the proportion of trading volume that involve the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Local* is the proportion of trading volume with the participation of *Locals* (traders classified as CTI 1 traders in the CFTC database). All figures are aggregated using trades time-stamped during business hours.

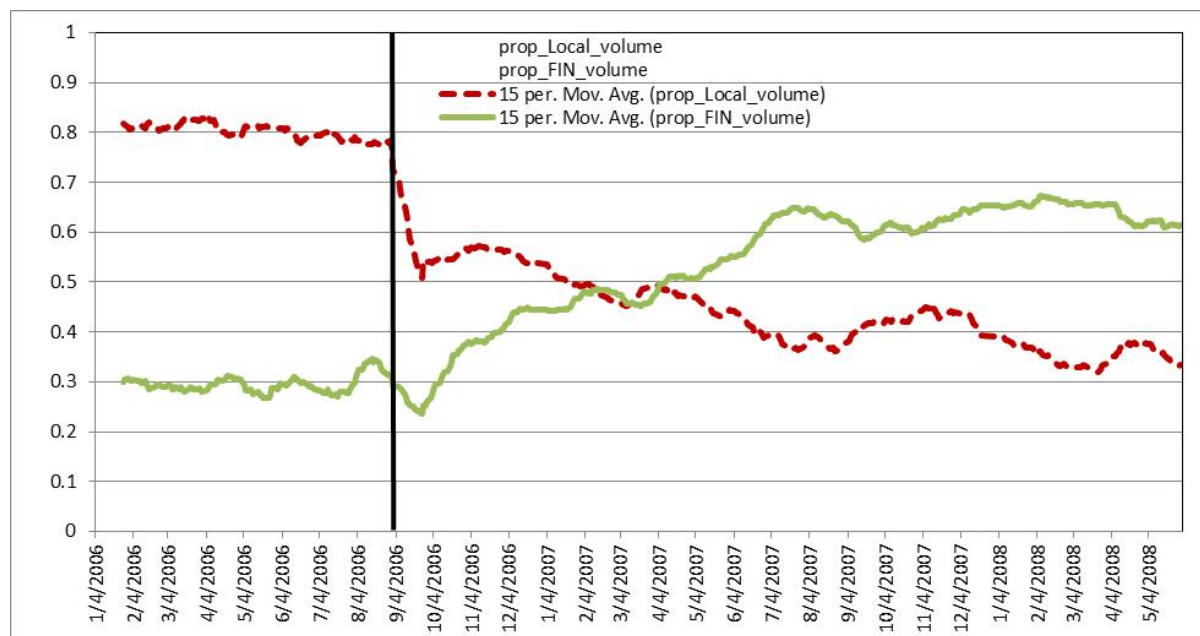
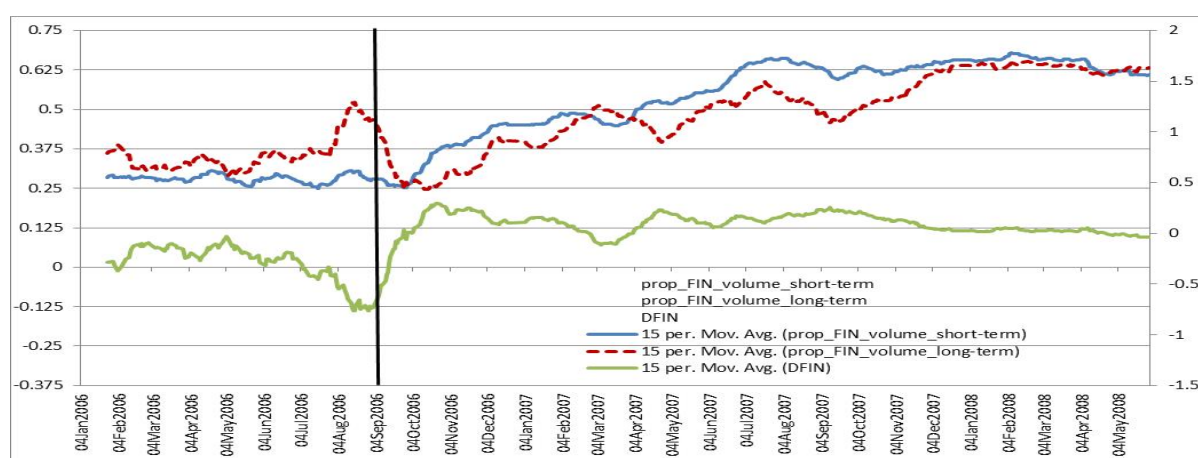


Figure 3: Financial Trading Activity: Short-Term vs. Long-Term contracts

Figure 3 compares the evolution of financial trading activity in short-term vs. long-term crude oil contracts. Panel A focuses on the proportion of trading volume involving financial traders in short-term vs. long-term crude oil contracts; and Panel B compares the number of new financial institutional traders in short-term vs. long-term crude oil contracts. The analysis is conducted on WTI sweet crude oil futures trading during business hours in the Pits for the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to May 31st, 2008). *FIN* is the proportion of trading volume with the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *FIN_Short-Term* is the daily, trading volume-weighted average of *FIN* across short-term contracts (contracts with less than 62 days to expiration). *FIN_Long-Term* is the daily, volume weighted average of *FIN* for long-term contracts (contracts with at least 62 days to expiration). ΔFIN is the daily percentage difference in the two: $(FIN_Short-Term - FIN_Long-Term) / FIN_Short-Term$. The red vertical line in the graphs identifies the date of the introduction of electronic trading – September 5th, 2006.

Panel A: Financial trading volume in short-term vs. long-term crude oil contracts



Panel B: Entry of new financial institutional traders in short-term vs. long-term crude oil contracts

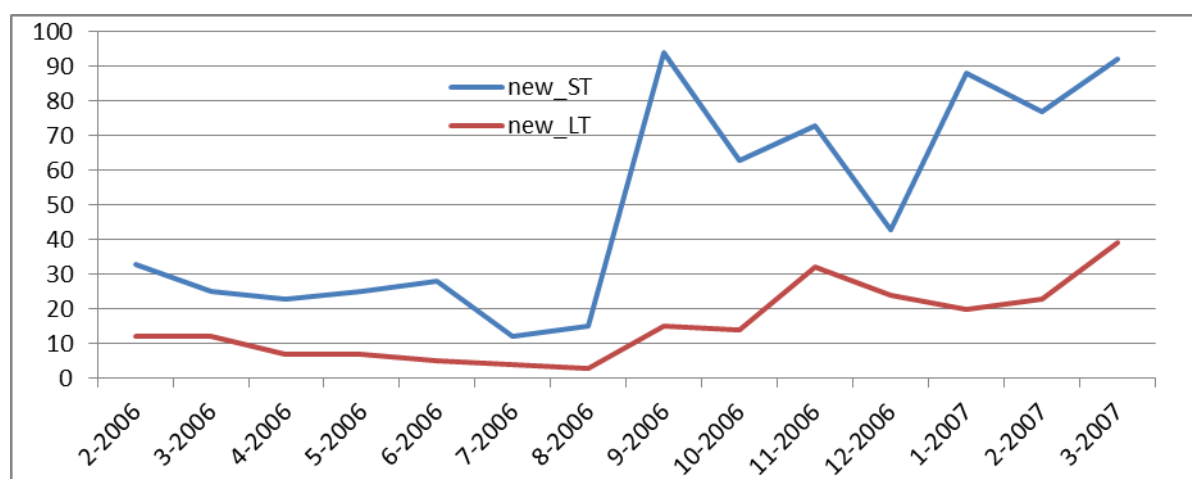
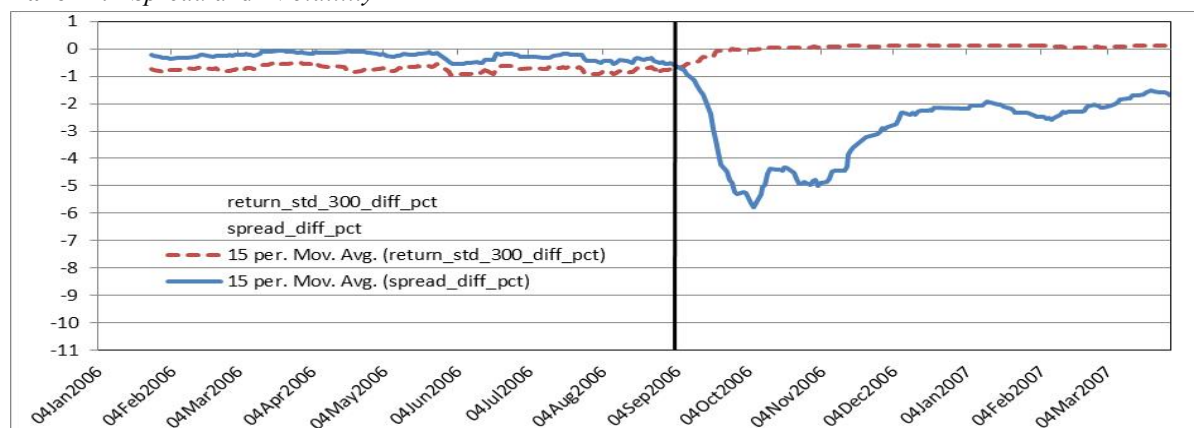


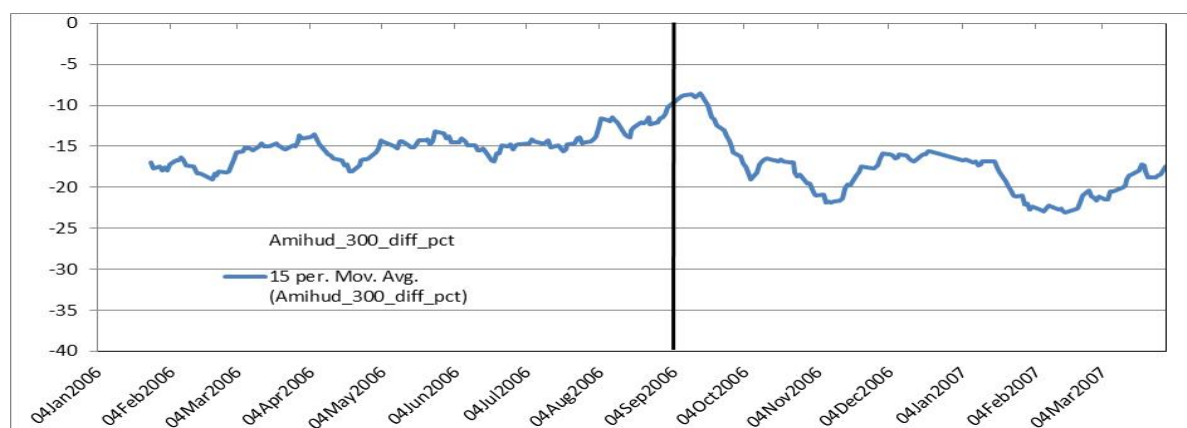
Figure 4: Financial Trading Activity & Market Quality measures: Short-Term vs. Long-Term contracts

Figure 5 compares the evolution of various market quality measures in short-term (contracts with less than 62 days to expiration) vs. longer-term crude oil contracts. The analysis is conducted for WTI sweet crude oil futures trading during business hours in the Pits for the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to May 31st, 2008). *Spread* is the daily average of 5 minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each interval (5 minutes), after classifying trades as buyer- vs. seller-initiated using the Lee and Ready (1991) tick test. *Amihud*, an inverse measure of depth (Amihud, 2002), is the daily average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. *AbsOIB* is the daily average of 5 minute customer (traders classified as CTI 4 traders in the CFTC database) trade imbalances calculated as the ratio of five-minute absolute trade imbalances (buyer-initiated *minus* seller-initiated trades) to trading volume. *PE_Variance* is the daily pricing error variance, estimated as in Hasbrouck (1993). *Price_Volatility* is the daily average of 5-minute volatility of intraday (log) transaction for each contract maturity in each interval. *PE_Proportion* is the daily ratio of *PE_Variance* and *Price_Volatility*. For each variable, Δ denotes the daily percentage difference in *Short-Term* vs *Long-Term* values. The red vertical line in each graph identifies the date of the introduction of electronic trading – September 5th, 2006.

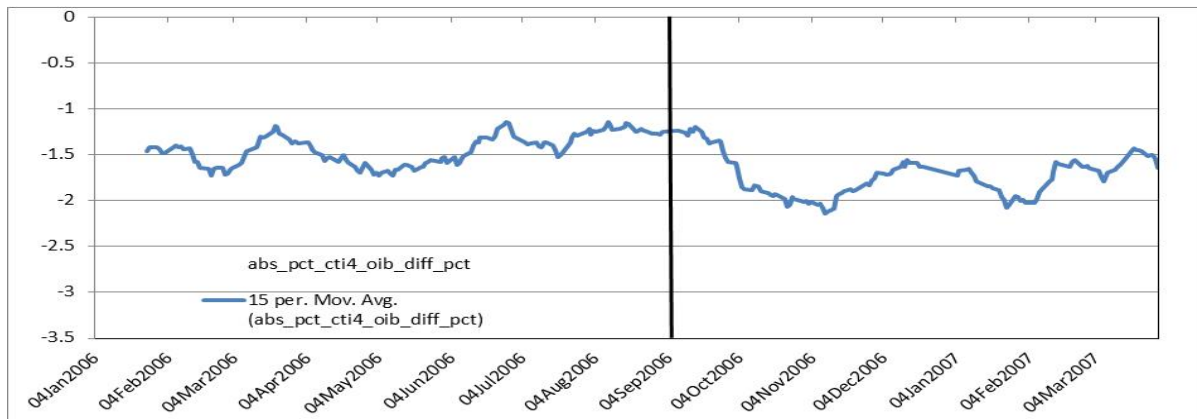
Panel A: $\Delta Spread$ and $\Delta Volatility$



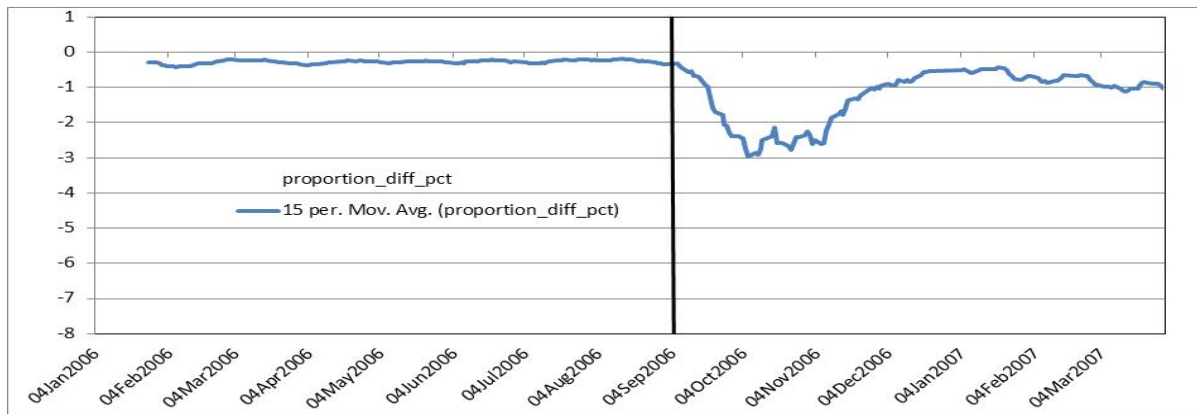
Panel B: $\Delta Amihud$



Panel C: $\Delta AbsOIB$



Panel D: $\Delta PE_Proportion$



Panel E: $\Delta PE_Variance$ and $\Delta Price_Volatility$

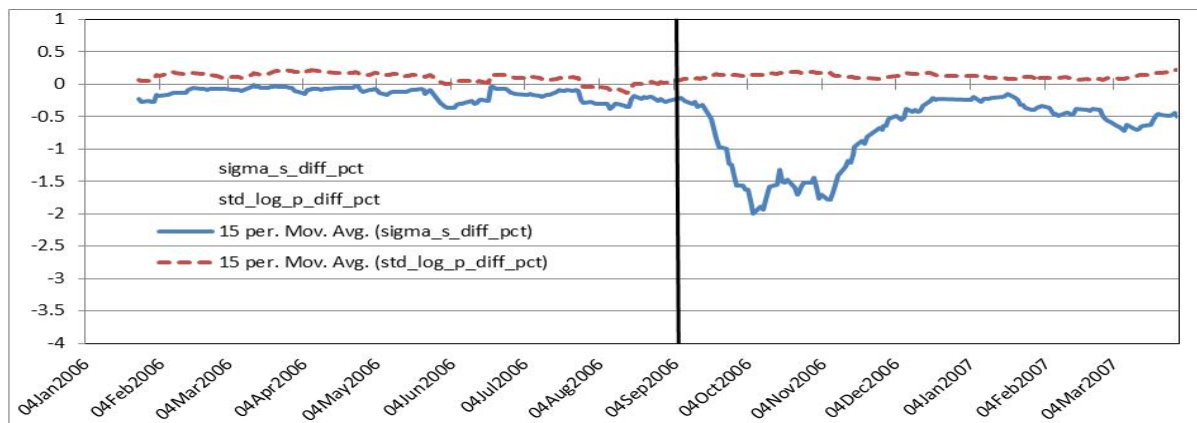
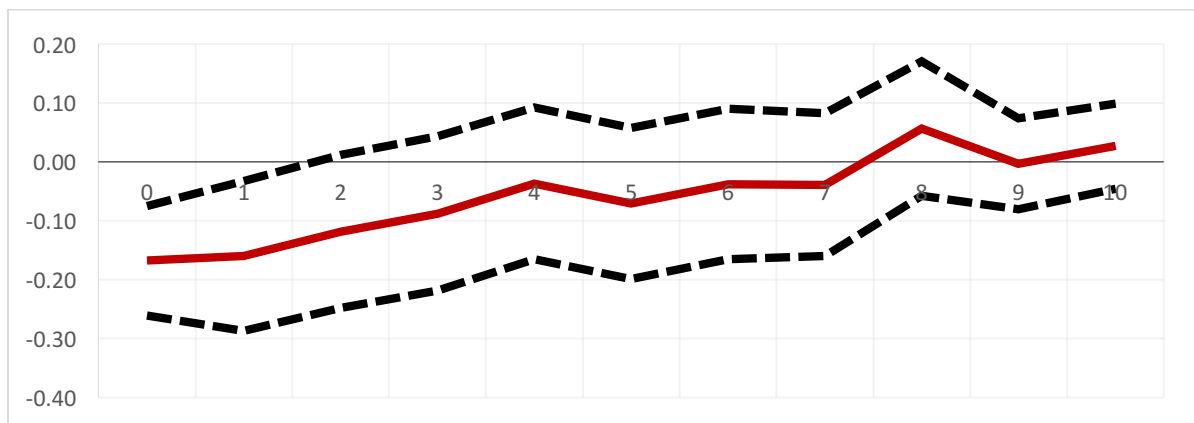


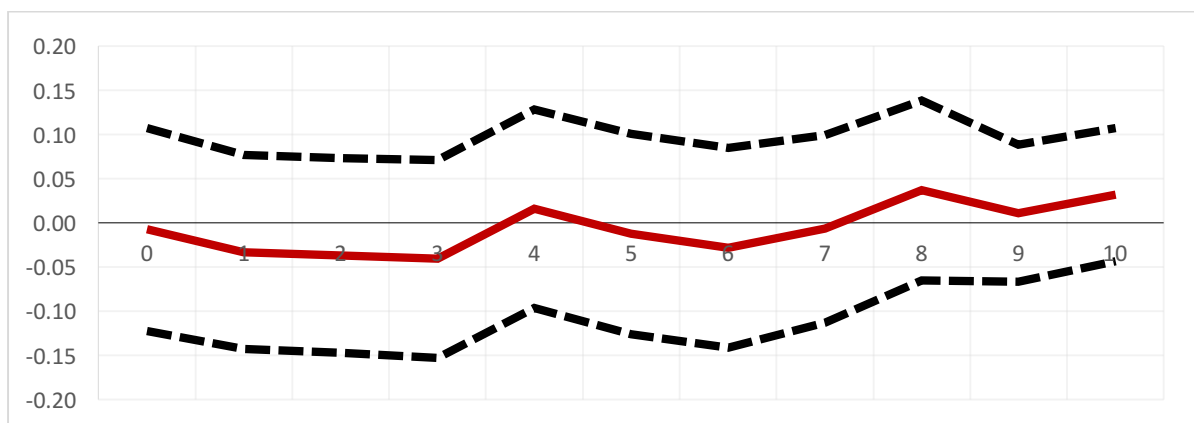
Figure 6 – Effect of Financial Traders: Impulse Response Functions

Figure 6 provides graphs of orthogonalized impulse response functions, depicting the impact of a one-standard deviation shock in the difference of the proportion of trading volume involving financial traders in short-term and long-term contracts on different measures of market quality in the WTI sweet crude-oil futures market. The analysis is conducted on the Globex platform from April 1st, 2007 to May 31st, 2008. *Spread* is the daily average of 5 minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each interval (5 minutes), after classifying trades as buyer- vs. seller-initiated using the tick-test. *Amihud*, an inverse measure of depth (Amihud, 2002), is the daily average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. *AbsOIB* is the daily average of 5-minute customer (traders classified as CTI 4 traders in the CFTC database) trade imbalances calculated as the ratio of five-minute absolute trade imbalances (buyer-initiated *minus* seller-initiated trades) to trading volume. *PE_Variance* is the daily pricing error variance, estimated as in Hasbrouck (1993). *Price_Volatility* is the daily average of 5-minute volatility of intraday (log) transaction for each contract maturity in each interval. *PE_Proportion* is the daily ratio of *PE_Variance* and *Price_Volatility*. *FIN* is the proportion of the trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *FIN_Short-Term* is the daily, trading volume-weighted average of *FIN* across short-term contracts (contracts with less than 62 days to expiration). *FIN_Long-Term* is the daily, volume weighted average of *FIN* across long-term contracts (contracts with greater than or equal to 62 days to expiration). ΔFIN is the daily percentage difference in the two ($(FIN_Short-Term - FIN_Long-Term) / FIN_Short-Term$). The remaining market quality variables are defined analogously. Impulse response functions are obtained for a VAR system consisting of 5 lags of ΔFIN , $\Delta Spread$, $\Delta Amihud$, and $\Delta AbsOIB$ and $\Delta PE_Proportion$.

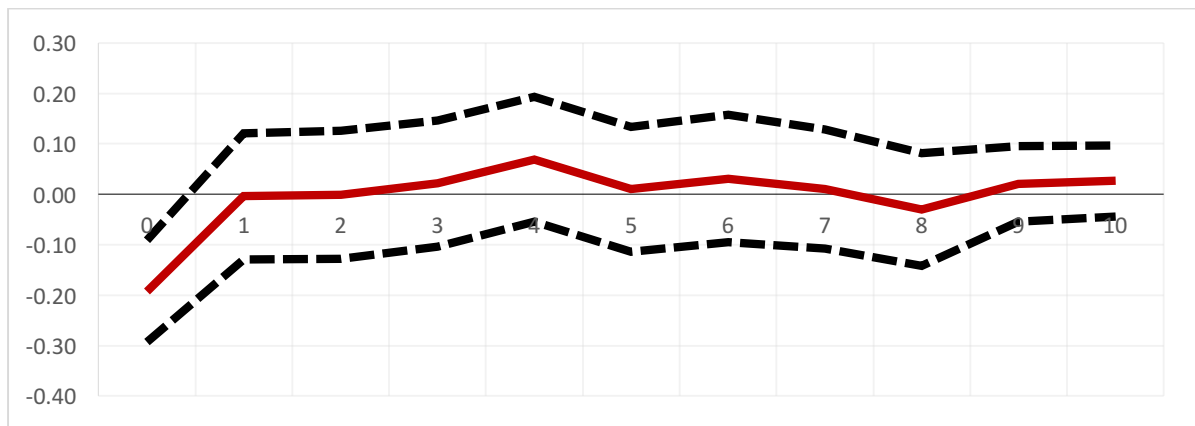
Panel A: Effect of ΔFIN on $\Delta Spread$



Panel B: Effect of ΔFIN on $\Delta Amihud$



Panel C: Effect of ΔFIN on $\Delta ABSOIB$



Panel D: Effect of ΔFIN on $\Delta PE_Proportion$

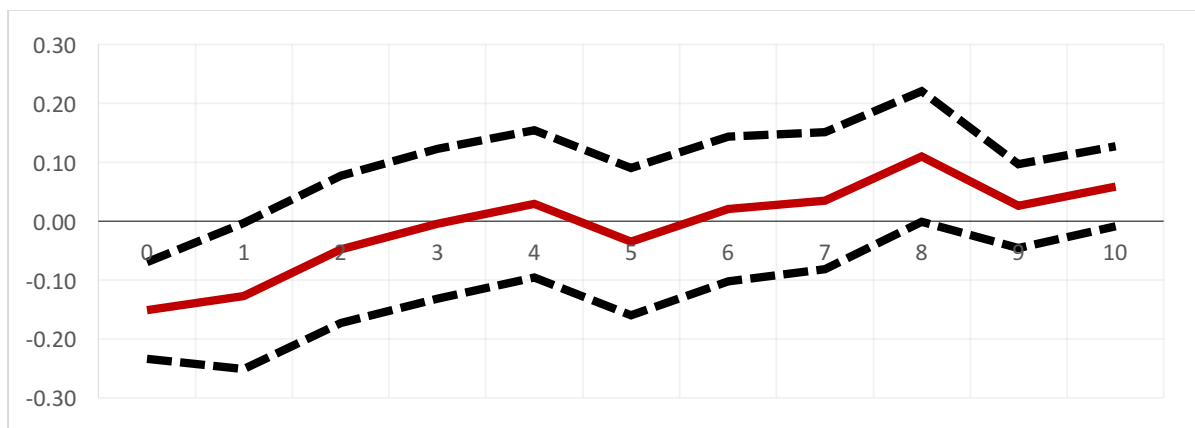


Table 1: Effect of Electronic Trading on Financial Traders and Market Quality – Event-Study

This table presents an analysis of financial traders' participation and measures of market quality surrounding the introduction of electronic trading (September, 2006). The sample period is January 3rd, 2006 to March 31st, 2007. *Pre-Electronification* refers to the period over January 3rd, 2006 to September 1st, 2006 and *Post-Electronification* refers to the period from September 5th, 2006 to March 31st, 2007. The analysis is conducted on WTI sweet crude-oil futures Pit trading in the *Pre-Electronification* period and on the Globex platform for the *Post-Electronification* period. *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Spread* is the daily average of 5 minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each interval (5 minutes), after classifying trades as buyer- vs. seller-initiated using the Lee and Ready (1991) tick-test. *Amihud*, an inverse measure of depth (Amihud, 2002), is the daily average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. *AbsOIB* is the daily average of 5 minute customer (traders classified as CTI 4 traders in the CFTC database) trade imbalances calculated as the ratio of five-minute absolute trade imbalances (buyer-initiated *minus* seller-initiated trades) to trading volume. *PE_Proportion* is the daily ratio of the pricing error variance, estimated as in Hasbrouck (1993), to the volatility of intraday (log) transaction prices. All the variables are estimated for each contract maturity and volume weighted averages are reported. Two tailed *p-values* are also reported.

	Pre- Electronification	Post- Electronification	Difference	Pct. Difference	p-value
FIN	29.64%	55.01%	25.37%	85.59%	<.001
Spread	0.37%	0.03%	-0.34%	-91.94%	<.001
Amihud	4.90	2.66	-2.24	-45.76%	0.604
AbsOIB	23.85%	13.48%	-10.37%	-43.48%	<.001
PE_Proportion	58.87%	3.73%	-55.14%	-93.66%	<.001

Table 2: Trader Description

This table describes various attributes of traders in the WTI sweet crude-oil futures market. The analysis presented in **Panel A** is conducted in the Pits during the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006); and **Panel B** presents the analysis conducted on the Globex platform over the *post*-electronification period (September 5th, 2006 to March 31st, 2007). *Locals*, *Fin. Inst.* and *Customers* are those classified as CTI 1, 2 and 4 traders respectively in the CFTC database. *Abs Closing Ratio* is the average ratio of a trader's ending-of-hour inventory and hourly trading volume. Similarly, *Trading Volume* and *Number of Trades* are also hourly averages of a trader's activity. Cross-sectional mean and median are also presented.

Panel A: Pits, between January 3rd, 2006 and September 1st, 2006

Traders	Trading Volume		Number of Trades		Abs Closing Ratio	
	Mean	Median	Mean	Median	Mean	Median
Locals	258	85	14	8	22%	3%
Fin. Inst.	269	81	6	3	57%	83%
Customers	87	10	4	2	75%	100%

Panel B: Globex, between September 5th, 2006 and March 31st, 2007

Traders	Trading Volume		Number of Trades		Abs Closing Ratio	
	Mean	Median	Mean	Median	Mean	Median
Locals	115	36	30	13	32%	12%
Fin. Inst.	298	50	114	19	38%	8%
Customers	77	10	23	4	60%	100%

Table 3: Effect of Electronic Trading on Financial Traders – Short-Term vs. Long-Term

This table presents an analysis of financial traders' participation and measures of market quality surrounding the introduction of electronic trading (September, 2006). The sample period is January 3rd, 2006 to March 31st, 2007. *Pre-Electronification* refers to the period over January 3rd, 2006 to September 1st, 2006 and *Post-Electronification* refers to the period from September 5th, 2006 to March 31st, 2007. The analysis is conducted on WTI sweet crude-oil futures Pit trading in the *Pre-Electronification* period and on the Globex platform for the *Post-Electronification* period. *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). All the variables are estimated for each contract maturity with volume weighted averages computed for (i) *Short-Term* (contract maturities with less than 62 days to expiration) and (ii) *Long-Term* (contract maturities with greater than or equal to 62 days to expiration) contracts separately. *FIN_Short-Term* is the daily, trading volume weighted average of *FIN* across *Short-Term* contracts. *FIN_Long-Term* is the daily, volume weighted average of *FIN* across *Long-Term* contracts. ΔFIN is the daily percentage difference in the two $((FIN_Short-Term - FIN_Long-Term) / FIN_Short-Term)$. Two tailed *p-values* are also reported.

	Pre-Electronification	Post-Electronification	Difference	p-value
FIN_Short-Term	28.10%	41.42%	13.32%	<.001
FIN_Long-Term	36.27%	37.48%	1.21%	0.312
ΔFIN	-30.42%	9.54%	39.96%	<.001

Table 4: Effect of Electronic Trading on Market Quality – Short-Term vs. Long-Term

This table presents an analysis of financial traders' participation and measures of market quality surrounding the introduction of electronic trading (September, 2006). The sample period is January 3rd, 2006 to March 31st, 2007. *Pre-Electronification* refers to the period over January 3rd, 2006 to September 1st, 2006 and *Post-Electronification* refers to the period from September 5th, 2006 to March 31st, 2007. The analysis is conducted on WTI sweet crude-oil futures Pit trading in the *Pre-Electronification* period and on the Globex platform for the *Post-Electronification* period. *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Spread* is the daily average of 5 minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each interval (5 minutes), after classifying trades as buyer- vs. seller-initiated using the Lee and Ready (1991) tick-test. *Amihud*, an inverse measure of depth (Amihud, 2002), is the daily average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. *AbsOIB* is the daily average of 5 minute customer (traders classified as CTI 4 traders in the CFTC database) trade imbalances calculated as the ratio of five-minute absolute trade imbalances (buyer-initiated *minus* seller-initiated trades) to trading volume. *PE_Proportion* is the daily ratio of the pricing error variance, estimated as in Hasbrouck (1993), to the volatility of intraday (log) transaction prices. All the variables are estimated for each contract maturity with volume weighted averages computed for (i) *Short-Term* (contract maturities with less than 62 days to expiration) and (ii) *Long-Term* (contract maturities with greater than or equal to 62 days to expiration) contracts separately. *Spread_Short-Term* is the daily, trading volume weighted average of *Spread* across *Short-Term* contracts. *Spread_Long-Term* is the daily, volume weighted average of *Spread* across *Long-Term* contracts. $\Delta Spread$ is the daily percentage difference in the two ($(Spread_Short-Term - Spread_Long-Term) / Spread_Short-Term$). The remaining market quality variables are defined analogously. Two tailed *p-values* are also reported.

	Pre-Electronification	Post-Electronification	Difference	p-value
Spread_Short-Term	0.35%	0.03%	-0.32%	<.001
Spread_Long-Term	0.41%	0.10%	-0.31%	<.001
$\Delta Spread$	-28%	-304%	-276%	<.001
Amihud_Short-Term	1.04	1.07	0.03	0.604
Amihud_Long-Term	15.00	20.00	5.00	<.001
$\Delta Amihud$	-14.87	-18.43	-3.56	<.001
AbsOIB_Short-Term	17.09%	12.36%	-4.73%	<.001
AbsOIB_Long-Term	40.85%	33.19%	-7.66%	<.001
$\Delta AbsOIB$	-143.74%	-175.26%	-31.52%	<.001
PE_Proportion_Short-Term	55.35%	4.12%	-51.23%	<.001
PE_Proportion_Long-Term	69.61%	8.90%	-60.71%	<.001
$\Delta PE_Proportion$	-27.73%	-132.27%	-104.54%	<.001

Table 5 – Effect of Electronic Trading on Financial Traders

This table shows the impact of the introduction of electronic trading on the difference in the rates of participation of financial traders in short-term vs. long-term contracts, and is also the first-stage of the instrumental variable regressions summarized in Table 3. The analysis is conducted for WTI sweet crude-oil futures trading in the Pits during the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform over the *post*-electronification period (September 5th, 2006 to March 31st, 2007). *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *FIN_Short-Term* is the daily, volume weighted average of *FIN* across Short-Term contracts (contracts with less than 62 days to expiration). *FIN_Long-Term* is the daily, volume weighted average of *FIN* across Long-Term contracts (contracts with greater than or equal to 62 days to expiration). ΔFIN is the daily percentage difference in the two $((FIN_Short-Term - FIN_Long-Term) / FIN_Short-Term)$, and is the ***dependent variable*** in the analysis. *Electronification* is a dummy variable equal to 1 in the *post*-electronification period. *EIA_Inventory* is a dummy variable set equal to 1 during EIA announcement days and 0 otherwise. *Lead_Inventory* is a dummy variable equal to 1 during the days prior to the EIA announcements. *First GSCI Roll* is a dummy variable equal to 1 on the five days when the monthly GSCI roll takes place and 0 otherwise. *Day of the Week* are a set of dummy variables for each day of the week. Two tailed *p-values* are also reported.

Parameter	Model 1		Model 2	
<i>Intercept</i>	-0.30	<.001	-0.08	0.508
<i>Electronification</i>	0.40	<.001	0.37	<.001
<i>VIX</i>			-0.02	0.018
<i>EIA_Inventory</i>			0.06	0.247
<i>Lead_Inventory</i>			0.01	0.868
<i>GSCI_Roll</i>			0.07	0.069
<i>Contract_Exp_Day</i>			-0.08	0.313
<i>Day of the Week</i>			YES	
<i>N</i>	299		299	
<i>Adj RSq</i>	30.65%		32.15%	

Table 6 – Effect of Financial Traders on Spreads

Table 3 presents analyses of the effect of the difference in the participation of financial traders in short-term and long-term contracts on the difference between the Bid-Ask spreads for short-term and long-term contracts. The analysis is conducted on WTI crude-oil futures trading in the Pits in the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform in the *post*-electronification period (September 5th, 2006 to March 31st, 2007). *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Volatility* is the daily volume-weighted average of the 5-minute volatility of (mid-quote) returns estimated for each contract in each maturity interval. *Volume* is the daily average of 5-minute trading volume estimated for each contract in each maturity interval. *Customer Volume* is the daily weighted average of the proportion of 5-minute trading volume involving customers (traders classified as CTI 4 traders in the CFTC database). *Spread* is the daily weighted average of 5 minute Bid-Ask spreads obtained using bid and asked prices estimated for each contract maturity in each interval (5 minutes), after classifying trades as buyer- vs. seller-initiated using the tick-test. *Spread_Short-Term* is the daily, volume weighted average of *Spread* across short-term contracts (contracts with less than 62 days to expiration). *Spread_Long-Term* is the daily, volume weighted average of *Spread* across long-term contracts (contracts with greater than or equal to 62 days to expiration). $\Delta Spread$ is the daily percentage difference in the two ($(Spread_Short-Term - Spread_Long-Term) / Spread_Short-Term$), and is the dependent variable in the analysis. ΔFIN , $\Delta Volatility$, $\Delta Volume$ and $\Delta Customer Volume$ are defined analogously. $\Delta FIN_Predicted$ is the fitted value of ΔFIN obtained from the first-stage regressions, presented in Table 2. *EIA_Inventory* is a dummy variable equal to 1 during EIA announcement days. *Lead_Inventory* is a dummy variable set equal to 1 during the days prior to the EIA announcements. *GSCI_Roll* is a dummy variable equal to 1 on the days of a GSCI roll. *Contract_Exp_Day* is a dummy variable equal to 1 on the day of the prompt contract's expiration. *Day of the Week* are a set of dummy variables for each day of the week. Two tailed *p-values*, obtained using Newey-West standard errors with 5 lags, are also reported.

Parameter	Model 1		Model 2		Model 3		Model 4		Model 5		Model 6	
<i>Intercept</i>	-0.95	<.001	1.57	0.019	-1.06	<.001	0.91	0.089	0.89	0.088	1.56	0.016
$\Delta FIN_Predicted$	-4.22	<.001	-3.64	<.001	-4.36	<.001	-3.89	<.001	-3.89	<.001	-3.64	<.001
$\Delta Volume$			-3.13	<.001			-2.42	<.001	0.80	<.001	-3.12	<.001
$\Delta Customer Volume$									-2.40	<.001	-0.05	0.912
$\Delta Volatility$	0.86	<.001	0.81	<.001	0.83	<.001	0.80	<.001	-0.11	0.823	0.81	<.001
<i>EIA_Inventory</i>	0.26	0.165	0.26	0.154	0.28	0.132	0.28	0.129	0.28	0.129	0.26	0.151
<i>Lead_Inventory</i>	0.00	0.998	0.02	0.884	0.02	0.901	0.03	0.824	0.03	0.835	0.02	0.890
<i>GSCI_Roll</i>	0.48	<.001	0.44	0.001	0.53	<.001	0.49	<.001	0.49	0.001	0.44	0.001
<i>Contract_Exp_Day</i>	-0.05	0.788	-0.11	0.547	-0.08	0.689	-0.11	0.513	-0.11	0.543	-0.10	0.570
<i>September, 2006</i>					-1.10	0.041	-0.94	0.079	-0.94	0.080		
<i>Dependent Lags</i>	2		2		2		2		2		2	
<i>Day of the Week</i>	YES		YES		YES		YES		YES		YES	
<i>N</i>	299		299		299		299		299		299	
<i>Adj RSq</i>	73.71%		74.75%		75.25%		75.81%		76.80%		75.69%	

Table 7 – Effect of Financial Traders on Depth

Table 4 presents analysis of the effect of the difference in the participation rates of financial traders in short-term vs. long-term contracts on differences in market depth for short-term vs. long-term contracts. The analysis is conducted on WTI sweet crude-oil futures trading in the Pits during the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to March 31st, 2007). *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Volatility* is the daily average of 5-minute volatility of (mid-quote) returns estimated for each contract in each maturity interval. *Volume* is the daily volume-weighted average of 5-minute trading volume estimated for each contract in each maturity interval. *Customer Volume* is the daily average of the proportion of 5-minute trading volume involving customers (traders classified as CTI 4 traders in the CFTC database). *Amihud*, an inverse measure of depth (Amihud, 2002), is the daily average of the ratio of absolute return to volume calculated in 5-minute non-overlapping intervals throughout the trading day. *Amihud_Short-Term* is the daily, volume weighted average of *Amihud* across short-term contracts (contracts with less than 62 days to expiration). *Amihud_Long-Term* is the daily, volume weighted average of *Amihud* across long-term contracts (contracts with greater than or equal to 62 days to expiration). $\Delta Amihud$ is the daily percentage difference between the two ($(Amihud_Short-Term - Amihud_Long-Term) / Amihud_Short-Term$), and is the **dependent variable** in the analysis. ΔFIN , $\Delta Volatility$, $\Delta Volume$ and $\Delta Customer Volume$ are defined analogously. $\Delta FIN_Predicted$ is the fitted value of ΔFIN obtained from the first-stage regressions, presented in Table 2. *EIA_Inventory* is a dummy variable equal to 1 during EIA announcement days. *Lead_Inventory* is a dummy variable equal to 1 during the days prior to the EIA announcements. *GSCI_Roll* is a dummy variable equal to 1 on the days of a weekly GSCI roll. *Contract_Exp_Day* is a dummy variable equal to 1 on the day of the prompt contract's expiration. *Day of the Week* are a set of dummy variables for each day of the week. Two tailed *p-values*, obtained using Newey-West standard errors with 5 lags, are also reported.

<i>Parameter</i>	Model 1		Model 2		Model 3		Model 4		Model 5		Model 6	
<i>Intercept</i>	-8.09	<.001	6.66	0.059	-9.01	<.001	9.15	0.010	7.97	0.024	5.51	0.118
<i>ΔFIN_Predicted</i>	-16.17	<.001	-11.00	<.001	-17.42	<.001	-11.76	<.001	-11.34	<.001	-10.60	<.001
<i>ΔVolume</i>			-17.24	<.001			-22.03	<.001	-5.38	0.022	-15.99	<.001
<i>ΔCustomer Volume</i>									5.67	<.001	4.92	0.038
<i>ΔVolatility</i>	5.91	<.001	5.62	<.001	5.95	<.001	5.63	<.001	-20.81	<.001	5.66	<.001
<i>EIA_Inventory</i>	0.28	0.736	0.21	0.799	0.32	0.689	0.27	0.732	0.07	0.928	0.02	0.976
<i>Lead_Inventory</i>	-0.72	0.395	-0.62	0.441	-0.71	0.397	-0.58	0.461	-0.66	0.406	-0.70	0.393
<i>GSCI_Roll</i>	-0.29	0.678	-0.72	0.315	-0.26	0.711	-0.78	0.247	-0.62	0.366	-0.57	0.433
<i>Contract_Exp_Day</i>	-2.66	0.019	-2.71	0.030	-2.78	0.011	-2.94	0.014	-2.53	0.033	-2.32	0.060
<i>September, 2006</i>					2.63	0.096	4.61	0.008	4.76	0.007		
<i>Dependent Lags</i>	2		2		2		2		2		2	
<i>Day of the Week</i>	YES		YES		YES		YES		YES		YES	
<i>N</i>	299		299		299		299		299		299	
<i>Adj RSq</i>	38.23%		41.81%		39.04%		44.45%		45.35%		42.52%	

Table 8 – Effect of Financial Traders on Customer Order Imbalances

Table 5 presents an analysis of the effect of the difference in the participation rates of financial traders in short-term vs. long-term contracts on differences in customer (relative) order imbalances for short-term vs. long-term contracts. The analysis is conducted on WTI sweet crude-oil futures trading in the Pits for the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to March 31st, 2007). *Volatility* is the daily average of 5-minute volatility of (mid-quote) returns estimated for each contract in each maturity interval. *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Volume* is the daily volume-weighted average of 5-minute trading volume estimated for each contract in each maturity interval. *Customer Volume* is the daily weighted average of the proportion of 5-minute trading volume involving customers (traders classified as CTI 4 traders in the CFTC database). *AbsOIB* is the daily average of 5-minute customer (traders classified as CTI 4 traders in the CFTC database) trade imbalances calculated as the ratio of five-minute absolute trade imbalances (buyer-initiated minus seller-initiated trades) to trading volume. *AbsOIB_Short-Term* is the daily, volume weighted average of *AbsOIB* across short-term contracts (contracts with less than 62 days to expiration). *AbsOIB_Long-Term* is the daily, volume weighted average of *AbsOIB* across long-term contracts (contracts with greater than or equal to 62 days to expiration). $\Delta AbsOIB$ is the daily percentage difference in the two ($(AbsOIB_Short-Term - AbsOIB_Long-Term) / AbsOIB_Short-Term$), and is the *dependent variable* in the analysis. ΔFIN , $\Delta Volatility$, $\Delta Volume$ and $\Delta Customer Volume$ are defined analogously. $\Delta FIN_Predicted$ is the fitted value of ΔFIN obtained from the first-stage regressions, presented in Table 2. *EIA_Inventory* is a dummy variable equal to 1 during EIA announcement days. *Lead_Inventory* is a dummy variable equal to 1 during the days prior to the EIA announcements. *First GSCI Roll* is a dummy variable equal to 1 on the day of a GSCI roll. *GSCI_Roll_Days2to4* is a dummy variable equal to 1 between the 2nd and 4th day after a GSCI roll. *Contract_Exp_Day* is a dummy variable equal to 1 on the day of the prompt contract's expiration. *Day of the Week* are a set of dummy variables for each day of the week. Two tailed *p-values*, obtained using Newey-West standard errors with 5 lags, are also reported.

<i>Parameter</i>	Model 1		Model 2		Model 3		Model 4		Model 5		Model 6	
<i>Intercept</i>	-1.20	<.001	-1.31	<.001	-1.30	<.001	-1.16	<.001	-0.85	0.009	-0.98	0.003
$\Delta FIN_Predicted$	-0.72	<.001	-0.76	<.001	-0.85	<.001	-0.81	<.001	-0.94	<.001	-0.90	<.001
$\Delta Volume$			0.14	0.742			-0.17	0.682	-0.52	0.183	-0.25	0.540
$\Delta Customer Volume$									1.54	<.001	1.55	<.001
$\Delta Volatility$	0.06	0.332	0.06	0.325	0.07	0.263	0.06	0.299	0.05	0.368	0.04	0.395
<i>EIA_Inventory</i>	-0.18	0.043	-0.17	0.044	-0.17	0.056	-0.17	0.056	-0.11	0.163	-0.12	0.135
<i>Lead_Inventory</i>	-0.13	0.123	-0.13	0.122	-0.13	0.138	-0.13	0.146	-0.10	0.208	-0.11	0.178
<i>GSCI_Roll</i>	0.17	0.007	0.18	0.007	0.18	0.004	0.18	0.005	0.13	0.031	0.12	0.040
<i>Contract_Exp_Day</i>	-0.20	0.113	-0.20	0.111	-0.03	0.758	-0.21	0.088	-0.33	0.005	-0.32	0.007
<i>September, 2006</i>					0.27	0.047	0.28	0.045	0.25	0.084		
<i>Dependent Lags</i>	2		2		2		2		2		2	
<i>Day of the Week</i>	YES		YES		YES		YES		YES		YES	
<i>N</i>	299		299		299		299		299		299	
<i>Adj RSq</i>	17.61%		17.35%		18.72%		18.48%		30.01%		29.10%	

Table 9 – Effect of Financial Traders on Pricing Efficiency

Table 6 presents analysis of the effect of the difference in the participation rates of financial traders in short-term vs. long-term contracts on differences in the pricing efficiency of short-term vs. long-term contracts. The analysis is conducted on WTI sweet crude-oil futures trading in the Pits for the *pre*-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the *post*-electronification period (September 5th, 2006 to March 31st, 2007). *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *Price_Volatility* is the daily volume-weighted average of 5-minute volatility of intraday (log) transaction for each contract in each maturity interval. *Volume* is the daily weighted average of 5-minute trading volume estimated for each contract in each maturity interval. *Customer Volume* is the daily weighted average of the proportion of 5-minute trading volume involving customers (traders classified as CTI 4 traders in the CFTC database). *PE_Variance* is the daily pricing error variance, estimated as in Hasbrouck (1993). *PE_Variance_Short-Term* is the daily, volume weighted average of *PE_Variance* short-term contracts (contracts with less than 62 days to expiration). *PE_Variance_Long-Term* is the daily, volume weighted average of *PE_Variance* across long-term contracts (contracts with greater than or equal to 62 days to expiration). $\Delta PE_Variance$ is the daily percentage difference between the two $((PE_Variance_Short-Term - PE_Variance_Long-Term) / PE_Variance_Short-Term)$, and is the **dependent variable** in the analysis. ΔFIN , $\Delta Price_Volatility$, $\Delta Volume$ and $\Delta Customer_Volume$ are defined analogously. $\Delta FIN_Predicted$ is the fitted value of ΔFIN obtained from the first-stage regressions, presented in Table 2. *EIA_Inventory* is a dummy variable equal to 1 during EIA announcement days. *Lead_Inventory* is a dummy variable equal to 1 during the days prior to the EIA announcements. *GSCI_Roll* is a dummy variable equal to 1 on days when the monthly GSCI roll takes place. *Contract_Exp_Day* is a dummy variable equal to 1 on the day of the prompt contract's expiration. *Day of the Week* are a set of dummy variables for each day of the week. Two tailed *p-values*, obtained using Newey-West standard errors with 5 lags, are also reported.

<i>Parameter</i>	Model 1		Model 2		Model 3		Model 4		Model 5		Model 6	
<i>Intercept</i>	-0.68	<.001	0.75	0.072	-0.65	<.001	0.42	0.338	0.37	0.397	0.70	0.086
$\Delta FIN_Predicted$	-1.03	<.001	-0.64	0.002	-0.92	<.001	-0.64	0.001	-0.61	0.003	-0.60	0.004
$\Delta Volume$			-1.71	0.001			-1.30	0.019	-1.23	0.025	-1.64	0.001
$\Delta Customer_Volume$									-0.33	0.244	-0.34	0.231
$\Delta Price_Volatility$	1.30	<.001	1.28	<.001	1.29	<.001	1.27	<.001	1.26	<.001	1.27	<.001
<i>EIA_Inventory</i>	0.01	0.953	0.01	0.926	0.01	0.908	0.01	0.896	0.00	0.971	0.00	0.998
<i>Lead_Inventory</i>	0.20	0.107	0.21	0.088	0.21	0.086	0.21	0.078	0.21	0.086	0.20	0.098
<i>GSCI_Roll</i>	0.24	0.001	0.21	0.001	0.24	0.001	0.22	0.001	0.23	0.001	0.22	0.001
<i>Contract_Exp_Day</i>	-0.20	0.179	-0.21	0.153	-0.18	0.144	-0.19	0.129	-0.16	0.195	-0.18	0.211
<i>September, 2006</i>					-0.53	0.034	-0.43	0.115	-0.42	0.115		
<i>Dependent Lags</i>	2		2		2		2		2		2	
<i>Day of the Week</i>	YES		YES		YES		YES		YES		YES	
<i>N</i>	299		299		299		299		299		299	
<i>Adj RSq</i>	35.32%		37.20%		37.60%		38.49%		38.52%		37.23%	

Table 10s – Effect of Non-Fast Financial Traders on Market Quality

Table 7 presents analysis of the effect of the difference in the participation rates of Non-Fast financial traders in short-term vs. long-term contracts on differences in the pricing efficiency of short-term vs. long-term contracts. The analysis is conducted on WTI sweet crude-oil futures trading in the Pits for the pre-electronification period (January 3rd, 2006 to September 1st, 2006) and on the Globex platform for the post-electronification period (September 5th, 2006 to March 31st, 2007). *FIN* is the proportion of trading volume involving the participation of financial traders (traders classified as CTI 2 traders in the CFTC database). *FIN_Non_FLP* is the proportion of non-fast financial traders: *FIN* – proportion of fast, financial liquidity providers (We identify FLPs as traders who trade more than 2000 trades a day and carry less than 5% of their daily trading volume overnight as in Raman, Robe and Yadav, 2014). *FIN Volatility* is the daily average of 5-minute volatility of (mid-quote) returns estimated for each contract in each maturity interval. *Volume* is the daily volume-weighted average of 5-minute trading volume estimated for each contract in each maturity interval. *Price_Volatility* is the daily volume-weighted average of 5-minute volatility of intraday (log) transaction for each contract in each maturity interval. *Volume* is the daily weighted average of 5-minute trading volume estimated for each contract in each maturity interval. *Customer Volume* is the daily weighted average of the proportion of 5-minute trading volume involving customers (traders classified as CTI 4 traders in the CFTC database). *PE_Variance* is the daily pricing error variance, estimated as in Hasbrouck (1993). *PE_Variance_Short-Term* is the daily, volume weighted average of *PE_Variance* short-term contracts (contracts with less than 62 days to expiration). *PE_Variance_Long-Term* is the daily, volume weighted average of *PE_Variance* across long-term contracts (contracts with greater than or equal to 62 days to expiration). $\Delta PE_Variance$ is the daily percentage difference between the two $((PE_Variance_Short-Term - PE_Variance_Long-Term) / PE_Variance_Short-Term)$. ΔFIN , $\Delta Price_Volatility$, $\Delta Volume$ and $\Delta Customer Volume$ are defined analogously. $\Delta FIN_Predicted$ is the fitted value of ΔFIN obtained from the first-stage regressions, presented in Table 2. *EIA_Inventory* is a dummy variable equal to 1 during EIA announcement days. *Lead_Inventory* is a dummy variable equal to 1 during the days prior to the EIA announcements. *GSCI_Roll* is a dummy variable equal to 1 on days when the monthly GSCI roll takes place. *Contract_Exp_Day* is a dummy variable equal to 1 on the day of the prompt contract's expiration. *Day of the Week* are a set of dummy variables for each day of the week. Two tailed p-values, obtained using Newey-West standard errors with 5 lags, are also reported.

<i>Parameter</i>	$\Delta Spread$		$\Delta Amihud$		$\Delta AbsOIB$		$\Delta PE_Variance$	
<i>Intercept</i>	0.05	0.941	5.01	0.224	-1.15	0.002	0.24	0.596
<i>$\Delta FIN_Non_FLP_Predicted$</i>	-7.55	<.001	-23.88	<.001	-2.09	<.001	-1.28	0.004
<i>$\Delta Volume$</i>	0.67	<.001	5.36	<.001	0.03	0.603	-1.38	0.010
<i>$\Delta Customer Volume$</i>	-3.17	<.001	-23.17	<.001	-0.69	0.062	-0.36	0.208
<i>$\Delta Volatility$</i>	-0.20	0.686	-5.78	0.015	1.50	<.001		
<i>$\Delta Price Volatility$</i>							1.26	<.001
<i>EIA_Inventory</i>	0.66	0.003	1.31	0.115	0.00	0.995	0.07	0.585
<i>Lead_Inventory</i>	0.05	0.745	-0.58	0.464	-0.10	0.244	0.21	0.083
<i>GSCI_Roll</i>	1.11	<.001	1.45	0.087	0.31	<.001	0.34	0.000
<i>Contract_Exp_Day</i>	-0.55	0.016	-4.02	0.002	-0.46	0.000	-0.24	0.071
<i>September,2006</i>	-0.91	0.089	4.69	0.007	0.25	0.083	-0.42	0.113
<i>Dependent Lags</i>	2		2		2		2	
<i>Day of the Week</i>	YES		YES		YES		YES	
<i>N</i>	299		299		299		300	
<i>Adj RSq</i>	76.80%		44.75%		29.87%		38.27%	